INTERNATIONAL MONETARY FUND

Global Pension Reform Database 1952 – 2025

From Narrative to Data Using AI

Prepared by Yudong Rao and Boele Bonthuis

WP/26/236

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2026
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WORKING PAPER

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IMF Working Paper
Fiscal Affairs Department

Global Pension Reform Database 1952 – 2025: From Narrative to Data Using AI

Prepared by Yudong Rao and Boele Bonthuis

Authorized for distribution by Rodrigo Cerda
October 2026

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ABSTRACT: This paper introduces the Global Pension Reform Database (GPRD), the first dataset to provide systematic coverage of pension reforms across 142 advanced, emerging, and developing economies from 1952 to 2025. Leveraging artificial intelligence (AI) through a ring-fenced, locally hosted large language model (LLM), we extract and classify more than 4,000 pension reform episodes from the Economist Intelligence Unit and the U.S. Social Security Administration by type, timing, status, and generosity, enabling consistent cross-country and historical analysis. Stylized facts show that pension reform has shifted from an advanced-economy phenomenon to a global and recurring policy tool, often clustered around fiscal and macroeconomic crises. An empirical application demonstrates its analytical potential: in non-advanced economies, implemented pension reforms are associated with statistically significant and persistent declines in the public debt-to-GDP ratio, suggesting that such reforms can contribute to sustained fiscal consolidation.

JEL Classification Numbers:	H55, J26, H53, E62, C80
Keywords:	Artificial intelligence; Pension reform; Social security; Public pensions; Benefit adequacy; Fiscal sustainability; Aging population; Labor market effects; Policy diffusion; Reform timing; Cross-country analysis; Large language models; Global Pension Reform Database
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WORKING PAPERS

Global Pension Reform Database 1952 – 2025:

From Narrative to Data Using AI

Prepared by Yudong Rao and Boele Bonthuis[1]

[1] The authors thank Mauricio Soto and Davide Furceri for their guidance, and Rodrigo Cerda, Era Dabla-Norris, Mitali Das, Christoph Freudenberg, Tatiana Goriainova, Radhika Goyal, Mengfei Gu, Jincheng (Eric) Huang, and the participants of the FAD Seminar Series for valuable comments and support. The authors also thank Ward Romp for sharing the pension-reform database, which helped cross-check the LLM results for OECD countries. Yudong Rao, who is currently at Washington University in St. Louis and was an intern at the IMF, is additionally grateful to David Amaglobeli, Gaetano Antinolfi, Ana Babus, Philip Dybvig, Bikas Joshi, Daehaeng Kim, Eswar Prasad, and Kiichi Tokuoka for their generous support and supervision, both at the Fund and in academia. The views expressed in this paper are those of the authors and do not necessarily represent the views of the IMF, its Executive Board, or IMF management.

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Acronyms

GPRD ..... Global Pension Reform Database

EIU ..... Economist Intelligence Unit (EIU) Country Reports

OECD ..... Organization for Economic Co-operation and Development

PAYG ..... Pay-as-You-Go

GDP ..... Gross Domestic Product

DiD ..... Difference-in-Differences

CPI ..... Consumer Price Index

SSA ..... Social Security Administration's International Update

LLM ..... Large Language Model

I. Introduction

Demographic change and rising fiscal pressures have made pension reform a priority for policymakers in countries across all income groups. Advanced economies face the challenge of sustaining benefit adequacy amid aging populations, while emerging markets and developing countries strive to expand coverage while ensuring long-term fiscal sustainability. Learning from past reform experiences, both domestic and international, can strengthen the design and implementation of future reforms.

The April 2025 Fiscal Monitor highlights the scale of the challenge. Pension spending averages about 8 percent of GDP in advanced economies, 4 percent in emerging markets, and 1.7 percent in developing countries. By 2050, these shares are projected to rise to 10.1, 8.1, and 2.8 percent of GDP, respectively. Without reform, rising pension expenditures risk undermining fiscal sustainability and constraining space for other spending priorities in many countries.

Existing empirical work on pension reform is largely concentrated on advanced economies. Narrative-based datasets compiled by Beetsma, Klaassen, Romp, and van Maurik (2020), Romp and Beetsma (2023), and Romp (2024) document reforms in 23 OECD countries from 1970 to 2023, and a dataset compiled by Bi and Zubairy (2023) covers 10 OECD countries between 1961 and 2019. These studies have advanced understanding of reform design, timing, and political economy drivers but leave substantial gaps for emerging and developing countries.

This paper introduces the Global Pension Reform Database (GPRD) to address this gap. Covering 142 economies from 1952 to April 2025, the GPRD identifies and classifies over 4,000 reform episodes drawn from 25,000 *Economist Intelligence Unit* country reports and 260 *Social Security Administration International Updates*. Using a ring-fenced large language model, pension reform episodes are coded into 21 standardized variables capturing type, status, generosity, and timing of different statuses.[2]

The paper has three objectives. First, it documents the construction of the GPRD, including source selection, extraction methods, and validation. Second, it presents stylized facts on the evolution, timing, and generosity of reforms across income groups. Third, it illustrates the empirical potential of the GPRD through an application examining the relationship between reform implementation and public debt dynamics.

The rest of the paper is organized as follows. Section II reviews the literature on text-as-data methods, narrative policy identification, and pension reform. Section III describes the construction of the Global Pension Reform Database (GPRD), including sources, extraction methods, and validation. Section IV introduces the database and compares it with existing datasets. Section V presents stylized facts on reform trends, timing, and generosity. Section VI applies the GPRD to examine the relationship between reform implementation and public debt. Section VII concludes with key findings and policy implications.

[2] A ring-fenced, locally hosted LLM refers to an AI model deployed on dedicated local hardware in a secure, isolated environment, avoiding the transmission of source materials to external AI services.

II. Literature Review

Our study connects three literatures: text-as-data methods in economics, narrative policy identification and event databases, and the economics of pension reform and social protection. On methods, the field has moved from dictionary and topic-model approaches to transformer architectures and LLM-assisted workflows. Foundational tutorials by Gentzkow, Kelly, and Taddy (2019) and Grimmer and Stewart (2013) codified best practices for text measurement and validation in the social sciences. More recent surveys emphasize the algorithmic pipeline used by economists, including embeddings and transformers, and stress the primacy of careful validation when moving from text to structured measures. Ash and Hansen (2023) offer a consolidated "text algorithms in economics" framework; Ludwig, Mullainathan, and Rambachan (2025) discuss applied econometric guardrails for LLMs, and Korinek (2023) reviews how generative models can augment research while highlighting limits.

A parallel line of work shows the value of news- and document-based indicators for macro and finance. Seminal contributions include the Economic Policy Uncertainty (EPU) index (Baker, Bloom, and Davis, 2016), firm-level political risk from earnings calls (Hassan et al., 2019), the Geopolitical Risk (GPR) index (Caldara and Iacoviello, 2022), and the World Uncertainty Index (Ahir, Bloom, and Furceri, 2022). These papers illustrate how transparent coding of large corpora can deliver policy-relevant time series with strong external validity, an ethos that was adopted in building the GPRD.

Methodologically, our contribution aligns with the emerging literature that leverages transformer-based LLMs to extract, classify, and structure economic information at scale. Recent applications include LLM-enabled synthesis of economics papers and causal claims, as well as supervised/LLM hybrids for policy measurement. Juhász, Lane, Oehlsen, and Pérez (2025) develop a text-based industrial policy database using modern language models; Clayton, Coppola, Maggiori, and Schreger (2025) examine geoeconomic pressures across countries. These studies demonstrate how LLMs can move beyond dictionaries to capture nuanced policy actions, precisely the shift our pipeline implements for pension reforms.

Our approach also builds on the narrative-identification tradition in macroeconomics, which reads policy records to code discrete events. Romer and Romer (2010) construct exogenous tax changes from archival sources; Ramey (2011) and Cloyne (2013) provide narrative measures of government spending and U.K. tax changes, respectively. This paper extends this tradition to a global, multi-decade setting by automating extraction and coding rules with an auditable LLM workflow, enabling the scale that hand-read narrative methods cannot achieve.

Within pensions, our work complements databases and syntheses that document system design and reform pressures. OECD's *Pensions at a Glance* compiles cross-country indicators and reform notes for OECD and G20 economies; SSA's *Social Security Programs Throughout the World* and *International Update* provide high-frequency updates on changes to program rules worldwide; while the ISSA's *Social Security Around the World* country profiles collect current pension system rules and parameters. Prior research examines the political economy and drivers of reform (Beetsma et al., 2020; Romp and Beetsma, 2023; Romp, 2024) and traces policy design in advanced and emerging economies (Amaglobeli et al., 2019). Bi and Zubairy (2023) construct a narrative-based database of pension reform motivations, implementation timing, and phase-in dynamics for ten OECD countries, showing that immediate versus phased-in reforms can lead to markedly different labor market responses among those near retirement. Our contribution is to extend **pension reform** coverage beyond OECD economies and to employ a ring-fenced AI pipeline that automatically extracts and structures unstructured narrative sources into structured micro-records of pension reform episodes.

III. Data Construction

A. Sources of Information: EIU and SSA

The data for the Global Pension Reform Database (GPRD) are extracted from Economist Intelligence Unit (EIU) country reports and the Social Security Administration's (SSA) International Update. This section provides an overview of these sources and the rationale for their selection.

1. Economist Intelligence Unit Country Reports (EIU)

The Economist Intelligence Unit (EIU) produces country reports for 189 countries, focusing on political, economic, and trade developments and their impacts on national risk profiles, covering the period from 1952 to the present for most countries. These reports leverage a vast network of field and headquarters-based experts, with headquarters analysts contributing at least five to seven years of experience and overseeing two to three countries each, ensuring current and specialized insights. The EIU's report production adheres to a rigorous five-step process: drafting by field experts, refining and standardizing by headquarters analysts, a thorough review by senior staff, sub-editing for clarity and accuracy, and final production checks for presentation quality. This systematic approach guarantees the reports' relevance, consistency, and alignment with the EIU's global perspectives (Ahir, Bloom, and Furceri, 2022).

2. Social Security Administration's International Update (SSA)[3]

The Social Security Administration's International Update (SSA) has been a monthly publication from the SSA's Office of Retirement and Disability Policy since October 2003. It offers a thorough examination of the latest developments in both public and private pension systems across the globe. Aimed at a wide audience, including policymakers, academic researchers, and professionals in the field of social security, this publication serves as a crucial tool for understanding the evolving landscape of pension reforms, legislative changes, and policy trends worldwide.

The EIU and SSA as primary sources were selected because together they offer the broadest available geographic and temporal coverage of pension reform activity. The EIU has been providing consistent and frequent reports since 1952, encompassing a wide array of countries and delivering in-depth analysis across various sectors such as politics, economy, and social issues. Notably, pension reforms are frequently detailed within the "Political Scene" and "Economic Policy" sections. Post-2012, the EIU transitioned its country reports to focus on medium- to long-term forecasts with its "Five-Year Forecast Report," enhancing the emphasis on strategic insights and long-term trends, albeit with a reduced focus on immediate policy changes, including pension reforms. To complement this and ensure thorough coverage of more immediate pension reform developments, the SSA's International Update, offering monthly reports since October 2003, has been integrated into our database. The SSA's updates provide timely insights into global pension reform, making it an invaluable resource for capturing short-term policy shifts. An added advantage of relying on the SSA is its exclusive focus on pension reforms, which significantly reduces the likelihood of noise in the data and ensures a high level of specificity and relevance in the information gathered. Together, the EIU and SSA offer a complementary combination, offering broad coverage and consistent updates on pension reforms worldwide. This synergy enables our database to provide a comprehensive view of pension reform episodes, from advanced economies to emerging and developing countries. An exploration of alternative sources, including

[3] Download link: https://www.ssa.gov/policy/research.html?sort=date&type=International%20Update

IMF staff reports, the NATLEX database, the ISSA database, the LABREF database, and OECD publications, revealed limitations, such as a focus restricted to OECD countries or current pension regulations, and inconsistent coverage. Consequently, the EIU and SSA emerge as the most robust and consistent sources available, offering unparalleled longitudinal and geographical breadth in the documentation of pension reforms.

Utilizing the Economist Intelligence Unit (EIU) and the Social Security Administration (SSA) as primary sources for our database entails certain limitations. The EIU reports, while providing a broad sweep of analysis across various countries and sectors, including politics, economy, and social issues, are authored by a diverse set of analysts. Not all these analysts specialize in pension reforms, which can lead to variability in the depth of coverage. This variability sometimes results in the omission of nuanced details or the overlooking of marginal pension reforms. For instance, changes in pension indexation, a critical area of pension reform that involves adjusting pensions to account for inflation and other factors, are covered in depth by the database of Bi and Zubairy (2023) but might be underrepresented or absent in EIU analyses. Additionally, the EIU's reliance on periodic reports introduces limitations. It may not capture reforms that occur between publication periods, potentially overlooking significant changes due to the nature of its sampling. An example that illustrates this limitation is the reporting on automatic adjustments in pension budgets, such as the increase reported in Argentina in 2010, stating, "As of September 1st, pensions will rise by 17%, benefiting 5.5m pensioners." While significant, such increases are often part of regular indexation processes rather than discrete pension reforms. This distinction is crucial as it affects the interpretation of data and the understanding of pension reform dynamics. Conversely, the SSA provides a more focused lens on social security and pension reform. Its reports, which commence from 2003, offer detailed coverage exclusively on these topics. This focus ensures that the SSA's updates are highly relevant and less likely to conflate regular indexation with substantial pension reforms. However, the SSA's narrower temporal range serves as a complementary perspective to the EIU's extensive historical breadth.

In the GPRD, a ‘pension reform episode’ is defined as any change to the rules, structure, or funding of a pension scheme or system. To address the breadth of this definition, reforms are classified as either parametric (minor adjustments within the existing system) or structural (major changes to system architecture, including pillar transitions). If information is available, the type of parametric or systemic reform is extracted.

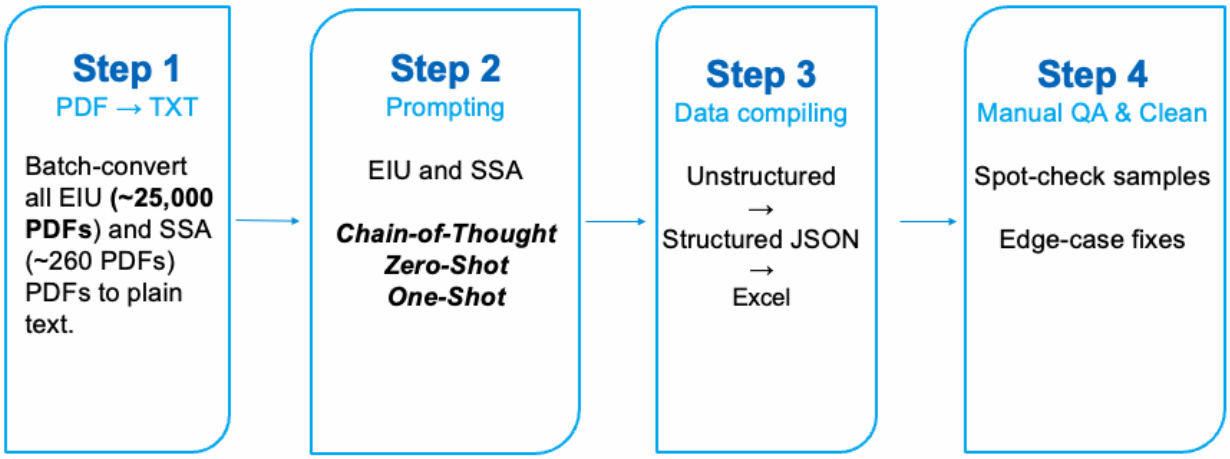
B. Methodology

This paper conducts large-scale extraction from two textual corpora, 25,000 EIU country reports and 260 SSA reports, using pre-trained, ring-fenced, open-weight large language models (LLMs). Specifically, it employs the 8-billion-parameter Llama 3.1-Instruct model released by Meta. To ensure compliance with EIU’s copyright requirements and to guarantee the reproducibility of our results, all extraction is deployed and executed locally using open-weight models with controlled pseudo-randomization. All resulting data are stored securely on local hardware, and the process runs continuously, 24 hours a day, for approximately seven days.

The selection of the Llama 3.1-8B-Instruct model reflects a balance between capability, computational efficiency, and adherence to data-handling constraints. As a free, open-weight model, it imposes no licensing costs, offering a clear advantage over proprietary API-based solutions. Its moderate size, 8 billion parameters, provides an optimal trade-off between accuracy and resource requirements, allowing faster inference and a smaller hardware footprint without compromising performance. This configuration enables timely completion of large-scale processing within the required seven-day window. Equally important, the model’s open-weight nature facilitates secure, ring-fenced self-hosting, thereby avoiding the external API calls required by proprietary API-based models, which are incompatible with EIU’s copyright policy.

C. Construction Steps

This paper implements the following four steps to transform narrative data from the EIU (~25,000 PDFs) and SSA (~260 PDFs) into a database of pension reforms:



Step 1: PDF to Plain-Text Conversion

All PDF documents were converted into plain-text format to enable processing by the LLM. A Python script using the PyPDF2 library batch-extracted text from the PDF reports, storing the output as UTF-8 encoded .txt files. The script iterated through each PDF, processed all pages sequentially, and logged any extraction errors without interrupting execution.

Step 2: LLM Extraction with Structured Prompts

All plain-text files were processed by the Llama 3.1-8B-Instruct model using a structured sequence of prompts. Before extracting information on pension reform measures, it was ensured that the LLM understood the concept of a pension reform episode in the context of this study. The model was instructed not to simply search for predefined keywords or rely on a static dictionary, since pension reforms are described differently across countries and documents. For example, *KiwiSaver* in New Zealand is a voluntary, work-based retirement savings scheme that may not explicitly mention the term *pension*, while Australian reports often use *superannuation* to refer to the public or occupational pension system.

No additional training was required because modern pre-trained LLMs, such as Llama 3.1-8B-Instruct, already have broad coverage of policy, economics, and pension-related terminology from their original training corpus. The model's instruction-tuned design enabled it to follow targeted prompts and identify relevant reform measures even when described using varied or region-specific terminology.

To maximize both accuracy and reproducibility, the prompts were implemented in a two-stage pipeline**:**

Stage 1 – Comprehension and Summarization: The model read the full plain-text document, applied the provided definitions and classification rules, and generated a concise structured summary of each detected reform episode in natural language.

Stage 2 – Structured Output Generation: The model converted the Stage 1 summary into a fully populated JSON record following a fixed schema of 31 fields. The schema includes categorical variables (e.g., reform type, status, reason), numerical variables (e.g., retirement ages, contribution rates), and free-text descriptors. JSON (JavaScript Object Notation) is a lightweight, machine-readable format that stores data as

name–value pairs, making the results both human-readable and easily parsed for database integration, automated validation, and statistical analysis.

This paper uses a hybrid prompt-engineering strategy designed to balance accuracy, generalizability, and reproducibility in the extraction process. The approach integrated four complementary elements. First, zero-shot extraction was used for substantive content recognition; no task-specific fine-tuning was performed, thereby maintaining the model's general applicability and avoiding any risk of data leakage. Second, few-shot classification cues were incorporated for selected categorical fields, such as Status of Pension Reform and Pension Reform Types, in the form of brief, generic examples intended to improve labelling consistency without introducing bias toward countries or periods. Third, one-shot format anchoring was implemented through the inclusion of a single real-world record (France 2023), used exclusively to lock the JSON schema formatting and field order while exerting no influence on the substantive extraction. Finally, hidden chain-of-thought reasoning was encouraged via explicit instructions for the model to "reason step-by-step" internally and to "re-check for completeness and logical consistency" prior to producing the final output; this was designed to improve accuracy while suppressing the display of intermediate reasoning in the output.

The complete text of both the Stage 1 and Stage 2 prompts, together with explanations of their structure and function, is presented below:

```
STEP1_PROMPT = r"""

You are an information extraction system assisting in reading and analyzing EIU
country reports and SSA reports to identify pension or social security reform measures
or episodes.

If EIU:
Your task: This file concerns a single country. Identify and extract all pension
reform episodes for that country only. Read the text in full, identify the pension
reform episodes, and answer the 31 structured questions below. Output must be in the
JSON format defined in STEP2_PROMPT.

If SSA:
Your task: This file may describe multiple countries. Identify every country for
which a pension reform is described with a status of Legislated, Planned, Announced,
Implemented, or Reversed (see definitions under Question 3), and ensure that no
country is omitted. Output must be in the JSON format defined in STEP2_PROMPT. For each
such country, perform a separate, complete pass through the text to answer all 1–31
questions before moving to the next country.
```

Then hidden chain-of-thought instructions were employed, which explicitly direct the model to reason internally and re-check its outputs, improving accuracy while keeping the reasoning out of the final output.

```
Before starting:
A. Read the text in full.
B. Internally reason step-by-step about each instruction and question, checking
   for explicit and implicit references.
C. Do not output your reasoning, only output the final JSON.
D. After producing the JSON, internally re-check for completeness, allowed values,
   date formats, and logical consistency.
```

Next, zero-shot prompting was applied to provide precise operational definitions of pension, pension scheme, pension system, and pension reform episodes/measures.

```
Definitions:
A. Pension: A retirement plan that provides regular payments after an individual
   stops working, usually at old age, or in cases of disability or widowhood.
B. Pension scheme: An arrangement to provide regular income in retirement, which
   can be occupational (employer-provided), private (insurance/bank), or public
   (government-provided).
C. Pension system: The national framework for providing pensions, potentially
   including multiple schemes.
D. Pension reform episodes/measures: Any change to the rules, structure, or
   funding of a pension scheme or system.
```

The Stage 1 prompt defined the model's role and adapted its approach to the document type. For EIU reports, which cover a single country, the model identified and summarized all pension reform episodes for that country. For SSA reports, which may cover multiple countries, the model first identified all countries with a recorded pension reform and then produced a separate summary for each country–episode pair. The resulting output, a structured natural-language summary covering all 31 database fields, was then passed to Stage 2 for conversion into JSON format.

```
### Extraction Questions and Embedded Instructions

1. Country Name
- Full country name or abbreviation. If absent, "Not explicitly mentioned".

2. Year of the article
- Publication year of the article/report. If absent, "Not explicitly mentioned".

3. Status of Pension Reform
- One or more of: Legislated, Planned, Announced, Implemented, Reversed, Other.
Separate multiple values with semicolons.
- Legislated = formally approved or bill passed.
- Planned = under discussion, not legislated.
- Announced = government communicated publicly, not legislated.
- Implemented = measures began taking effect.
- Reversed = undoing or abolishing a previous reform.
Examples:
     - "Government passes bill raising retirement age" → Legislated
     - "Cabinet proposes reform to be debated next month" → Planned
     - "President announces new pension plan" → Announced

4. legislation_date
- Date reform was legislated. mm-yyyy format if possible; else "Not explicitly
mentioned". If stated as “today” or “yesterday”, use those.

5. Planned_date
```

- Date reform was planned/discussed but not legislated. mm-yyyy format if possible; else "Not explicitly mentioned".

6. announcement_date
   - Date reform was announced. mm-yyyy format if possible; else "Not explicitly mentioned".

7. implementation_date
   - Date reform began to take effect. mm-yyyy format if possible; else "Not explicitly mentioned".

8. reversal_date
   - Date reform was reversed. mm-yyyy format if possible; else "Not explicitly mentioned".

9. Pension Reform Types
   - Assign one or more categories (separate with semicolons).
   - Categories:
     - indexation: changes to benefit adjustment rules (e.g., inflation-link).
     - coverage: changes to who is included/excluded.
     - benefit rule changes: changes to eligibility, accrual, replacement rates.
     - systematic (structural) pension reforms: major system redesigns.
     - funding mechanisms: changes to financing or contribution structure.
     - retirement age: changes to statutory retirement ages.
     - pension portability: ability to transfer pension rights between jobs/systems.
     - investment strategies: changes to pension fund investment policies.
     - administration and governance: management, oversight, transparency changes.
     - sustainability and solvency measures: ensure long-term financial health.
     - pension plan types: introduction/removal of plan types (DB, DC, hybrid).
     - eligibility and access: changes in who can join or claim.
     - others: relevant changes not in above categories.

   Examples:
     - "Link pensions to wage growth" → indexation
     - "Raise eligibility age from 62 to 64" → retirement age
     - "Introduce funded second pillar" → systematic (structural) pension reforms
     - If none, "Not explicitly mentioned".

10. indexation_change
    - One of: more generous indexation, less generous indexation, Not explicitly mentioned.

11. coverage_change
    - One of: extended coverage, reduced coverage, Not explicitly mentioned.

12. benefit_change

- One of: more generous benefits, less generous benefits, Not explicitly mentioned.

13. Social_insurance_change

- Yes, No, or Not explicitly mentioned. Refers to mandatory government-managed schemes covering most private workers.

14. Reform_category

- Parametric reform: Adjusts parameters such as retirement ages, contribution rates, benefit formulas, indexation, or coverage.
- Structural reform: Redesigns the pension system (e.g., new multi-pillar system, shift from DB to DC, introduction of NDC schemes).
- Not explicitly mentioned.

15. reform_coverage

- Targeted reform (specific groups), Broad-based reform (everyone), Mixed, or Not explicitly mentioned.

16. reason_reform

- Assign one or more categories (separate with semicolons): Economic growth, Fiscal reasons, Benefit adequacy, Distributional impact, Economic impact, Inflation, Demographic changes, Inequality and fairness, Labor market changes, Other, Not explicitly mentioned.

17. relevancetopr_score

- Float 0.0–10.0 rating relevance to pension reform.
- 0= not about pension reform at all
- 10 = highly relevant to pension reforms

18. retirement_age_change

- retirement age increased, retirement age decreased, Not explicitly mentioned. Must be legislated.

19. Retirement_age_change_details

- Details of affected ages and magnitude of change.

20. Retirement_age_pre_reform_female

- A string describing the retirement age(s) of female before the reform; separate multiple ages with semicolons.

- Must refer to a legislated change, meaning status in 3 contains "Legislated"

21. Retirement_age_after_reform_female

- A string describing the retirement age(s) of female after the reform; separate multiple ages with semicolons.

- Must refer to a legislated change, meaning status in 3 contains "Legislated"

22. Retirement_age_pre_reform_male

- A string describing the retirement age(s) of male before the reform; separate multiple ages with semicolons.
- Must refer to a legislated change, meaning status in 3 contains "Legislated"

23. Retirement_age_after_reform_male
- A string describing the retirement age(s) of male after the reform; separate multiple ages with semicolons.
- Must refer to a legislated change, meaning status in 3 contains "Legislated"

24. Contribution_rate_change
- contribution rate increased, contribution rate decreased, or Not explicitly mentioned. Must be legislated.

25. Old Employer Contribution Rate
- Rate before reform; "Not explicitly mentioned" if absent.
- Must refer to a legislated change, meaning status in 3 contains "Legislated"

26. New Employer Contribution Rate
- Rate after reform; "Not explicitly mentioned" if absent.
- Must refer to a legislated change, meaning status in 3 contains "Legislated"

27. Old Employee Contribution Rate
- Rate before reform; "Not explicitly mentioned" if absent.
- Must refer to a legislated change, meaning status in 3 contains "Legislated"

28. New Employee Contribution Rate
- Rate after reform; "Not explicitly mentioned" if absent.
- Must refer to a legislated change, meaning status in 3 contains "Legislated"

29. Old Contribution Rate (Employee and Employer Combined)
- Combined rate before reform; "Not explicitly mentioned" if absent.
- Must refer to a legislated change, meaning status in 3 contains "Legislated"

30. New Contribution Rate (Employee and Employer Combined)
- Combined rate after reform; "Not explicitly mentioned" if absent.
- Must refer to a legislated change, meaning status in 3 contains "Legislated"

31. Is the legislated pension reform minor or major?
- Minor, Major, or N/A.
- Major = large-scale, structural, long-term impact.
- Minor = small-scale, incremental.

Final instructions before output:

```
    - Read the document in full and reason step-by-step internally.
    - For each country and reform episode, compile a concise natural-language summary
covering the information needed to answer Questions 1–31.
    - If information is missing, note “Not explicitly mentioned” in the summary.
    - Do not output JSON. Return only the country- and episode-structured summary for
use in Step 2.
    - Internally re-check for completeness, allowed values, date formats, and logical
consistency before returning the summary.
    """
```

Following Stage 1, which produces structured natural-language summaries of each pension reform episode, Stage 2 transforms these summaries into database-ready JSON records. Using zero-shot schema rules, the prompt enforces the inclusion of all 31 database fields, specific placeholder values, and strict formatting standards. A one-shot format anchor (France 2023) is provided to lock field order, value formatting, and placeholder usage, reducing structural errors. Finally, hidden chain-of-thought self-checking instructs the model to internally verify completeness and schema compliance before returning results, ensuring the final output is machine-readable, consistent, and ready for integration with minimal post-processing.

```
    STEP2_PROMPT = r"""
    Now, based on your previous summary from Stage 1, convert it into one or more JSON
objects. Each object corresponds to one country and potentially one pension reform
“episode.” The Stage 2 prompt enforces strict formatting and schema compliance,
requiring that all 31 database fields are present and populated with either extracted
values or the specified default placeholders.

    Remember:
    - Internally re-check the output before returning.
    - All 31 fields must be present in each object.
    - Dates must be in mm-yyyy format or allowed strings ("today", "yesterday", "Not
explicitly mentioned").
    - Allowed categorical values must be used exactly.

    Example:
    [
      {
        "Country Name": "France",
        "Year of the article": "2023",
        "Status of Pension Reform": "Announced;Legislated",
        "legislation_date": "04-2023",
        "Planned_date": "Not explicitly mentioned",
        "announcement_date": "04-2023",
        "implementation_date": "Not explicitly mentioned",
        "reversal_date": "Not explicitly mentioned",
        "Pension Reform Types": "retirement age;benefit rule changes",
        "indexation_change": "Not explicitly mentioned",
        "coverage_change": "Not explicitly mentioned",
        "benefit_change": "Not explicitly mentioned",
        "Social_insurance_change": "Not explicitly mentioned",
```

```
        "Reform_category": "Parametric Reform",
        "reform_coverage": "Broad-based reform (applies to everyone)",
        "reason_reform": "Fiscal reasons",
        "relevancetopr_score": 10.0,
        "retirement_age_change": "retirement age increased",
        "Retirement_age_change_details": "62 to 64; applied to all",
        "Retirement_age_pre_reform_female": "62",
        "Retirement_age_after_reform_female": "64",
        "Retirement_age_pre_reform_male": "62",
        "Retirement_age_after_reform_male": "64",
        "Contribution_rate_change": "Not explicitly mentioned",
        "Old Employer Contribution Rate": "Not explicitly mentioned",
        "New Employer Contribution Rate": "Not explicitly mentioned",
        "Old Employee Contribution Rate": "Not explicitly mentioned",
        "New Employee Contribution Rate": "Not explicitly mentioned",
        "Old Contribution Rate (Employee and Employer Combined)": "Not explicitly
mentioned",
        "New Contribution Rate (Employee and Employer Combined)": "Not explicitly
mentioned",
        "Is the legislated pension reform minor or major?": "major"
      }
    ]

    Notes:
    - Multiple objects are allowed if multiple countries or episodes are in the same
input.
    - "Status of Pension Reform" can contain multiple values separated by semicolons.
    - Output must be strictly valid JSON with no text before or after.
    """
```

Step 3: Aggregating JSON Output into Tabular Format

The structured JSON outputs from Step 2 were converted into an analysis-ready table. A Python script flattened nested JSON data, preserved key metadata (e.g., filename, year, month), and exported the aggregated dataset as a UTF-8 encoded CSV compatible with Excel. This step standardized heterogeneous JSON outputs into a consistent rectangular dataset for downstream analysis.

Step 4: Manual Quality Assurance and Data Cleaning

The final stage implemented a structured QA and cleaning process to produce a reliable, analysis-ready dataset. First, all observations with a relevancetopr_score below 5 were removed to eliminate noisy and spurious cases. Second, records containing multiple pension reform statuses (e.g., Legislated and Planned) were split into separate rows, ensuring that each record contained only one status. Third, only records with statuses of Legislated, Announced, Implemented, or Reversed were retained, excluding Planned and other missing-status entries. Fourth, date variables were harmonized by ensuring that each status had a valid corresponding date; placeholders such as “today” or “yesterday” were replaced with the month and year extracted from the document filename. Fifth, the data were reshaped so that each status–date pair was stored in a single standardized column. Sixth, country names were standardized to ensure consistency across the EIU and SSA extractions. Seventh, a unique identifier was generated for each record by combining the country

name, date, and status, and duplicates with the same identifier were removed. Eighth, each variable was cleaned and checked for formatting and plausibility, such as verifying that retirement ages and contribution rates fell within reasonable ranges, to guard against spurious outputs caused by large language model "hallucinations." Finally, for OECD countries, entries were manually cross-checked against Bi and Zubairy (2023) and Romp (2024) to validate accuracy. As this process involved more than 250 million words spanning 142 countries, the full LLM extraction required multiple iterations over more than seven days to complete. In total, 4,056 pension reform episodes (measures) were captured. While extensive manual checks were performed, some errors may remain; the caveats and limitations are discussed in detail in the database introduction section.

IV. Database Introduction

A. Description

The Global Pension Reform Database (GPRD) is the first dataset to provide systematic, cross-country coverage of pension reform measures across both advanced and, crucially, emerging and developing economies. Built using ring-fenced AI, it spans 142 countries from 1952 to April 2025, with later implementation dates included for reforms with long legislative-to-execution timelines. By processing roughly 250 million words of narrative text through a fully automated and adaptable pipeline that is computationally reproducible conditional on access to the underlying corpora, this paper illustrates how large language models can convert unstructured policy narratives into structured, analysis-ready data in economic areas where coverage is sparse and uneven across countries. Inspired by and building on the valuable foundations of Beetsma et al. (2020), Bi and Zubairy (2023), and Romp (2024), the GPRD represents a methodological advance that extends systematic pension reform coverage far beyond OECD economies, enabling comprehensive analysis of reforms in regions that have historically been underrepresented in global datasets. The database includes the following variables:

- **Status**
  This variable indicates the stage of the pension reform episode—whether it is an *announcement*, *legislation*, *implementation*, or *reversal*. A single reform measure may be recorded at multiple stages with different corresponding dates. For example, in France, the pension reform that raised the minimum retirement age was signed into law in April 2023 (recorded as *legislation*), while its provisions came into effect in September 2023 (recorded as *implementation*).
- **Country**
  Denotes the official name of the country in which the pension reform occurred.
- **Country Code**
  The standardized three-letter country code, based on the ISO Alpha-3 format, used for consistent cross-country referencing.
- **Date**
  Captures the month and year associated with the reform episode, depending on its status. As the variable is standardized at the monthly level, specific dates (e.g., April 15, 2023) are recorded as the first of the month (e.g., April 1, 2023) in the format 4/1/2023 or Apr-23.

- **Pension Reform Types**

  This variable categorizes the content and focus of each reform measure. Reform types include:

  1. Indexation: Changes in how pensions are adjusted (e.g., switching from wage to price indexation).
  2. Coverage: Expansion or contraction of the groups eligible for pension benefits
  3. Benefit Rule Changes: Modifications in replacement rates, benefit formulas, or minimum guarantees.
  4. Systemic (Structural) Reforms: major shifts in pension system architecture (e.g., transitioning from PAYG to funded schemes).
  5. Funding Mechanisms: Changes in the financing of pension systems (e.g., new taxes or contributions).
  6. Retirement Age: Adjustments to statutory or early retirement ages
  7. Pension Portability: Reforms enhancing the ability to transfer pension rights across systems or employers.
  8. Investment Strategies: Changes in asset allocation rules or investment regulations
  9. Administration and Governance: reforms targeting the management, oversight, or efficiency of pension institutions.
  10. Sustainability and Solvency Measures: Reforms aiming to improve long-term financial health.
  11. Pension Plan Types: Changes in the nature or mix of defined benefit (DB), defined contribution (DC), or hybrid plans.
  12. Eligibility and Access: Alterations to qualifying conditions or vesting periods
  13. Other: Residual category for reforms not covered above.

A single reform episode may fall under multiple types simultaneously, depending on its scope and design.

- **Indexation Change**

  This variable captures whether the reform modified the indexation rule used to adjust pension benefits over time. It includes two main categories:

  1. "More Generous Indexation": Reforms that increased the generosity of indexation (e.g., shifting from price indexation to wage indexation or introducing ad hoc increases above inflation)
  2. "Less Generous Indexation": Reforms that reduced the generosity of indexation (e.g., moving from wage to price indexation, freezing benefits, or applying below-inflation adjustments)

- **Coverage Change**

  This variable indicates whether the pension reform affected the breadth of population coverage within the pension system. It includes two possible categories:

  1. "Extended Coverage": Reforms that expanded eligibility or enrollment, such as including informal sector workers, self-employed individuals, or uncovered population groups
  2. "Reduced Coverage": Reforms that narrowed eligibility, excluded specific groups, or introduced restrictions that limited access to pension benefits

- **Benefit Change**

  This variable captures whether the reform altered the generosity of pension benefits. It includes two main classifications:

  1. "More Generous Benefits": reforms that increased benefit levels, replacement rates, or introduced new benefit guarantees

2. "Less Generous Benefits": reforms that reduced benefit generosity, such as lowering replacement rates, tightening benefit formulas, or cutting minimum pension amounts

- **Social Insurance Change**

  This binary variable indicates whether the pension reform involved changes to the social insurance component of the pension system:

  1. "Yes" : if the reform modified the structure, financing, eligibility, or benefit rules of social insurance-based pension schemes (typically contributory public systems)
  2. "No" : if the reform did not affect the social insurance component

- **Reform Category**

  This variable classifies the nature of the pension reform based on its scope and depth:

  1. "Parametric Reform" refers to adjustments made within the existing pension system without altering its fundamental structure. Examples include changes to retirement age, benefit formulas, contribution rates, or indexation rules.
  2. "Structural Reform" denotes fundamental shifts in the design or institutional framework of the pension system. This may involve the introduction of a new multi-pillar system, transitions from pay-as-you-go to funded schemes, or the replacement of defined benefit plans with defined contribution plans.

- **Reform Coverage**

  This variable describes the scope of the population affected by the pension reform:

  1. "Targeted reform (specific sectors/groups)" refers to reforms that apply only to particular segments of the population, such as civil servants, military personnel, or informal sector workers.
  2. "Broad-based reform (applies to everyone)" captures reforms that are universally applied across all workers or beneficiaries, regardless of sector or employment status.
  3. "Mixed (some measures apply to specific sectors/groups and some to everyone)" is used when a reform package includes both universal and targeted components, affecting both broad and specific groups simultaneously.

- **Reason Reform**

  This variable identifies the stated or inferred motivations behind a pension reform. A single reform episode may be associated with multiple reasons simultaneously, reflecting the multidimensional drivers of pension policy changes. The categories include:

  1. "Economic Growth": Reforms aimed at enhancing long-term growth potential by improving labor incentives, savings, or capital market efficiency.
  2. "Fiscal Reasons": Reforms introduced to reduce pension-related expenditures or improve the sustainability of public finances.
  3. "Benefit Adequacy": Reforms intended to ensure sufficient retirement income, often through benefit increases or targeted support for vulnerable groups.
  4. "Distributional Impact": Reforms designed to improve equity across income groups, generations, or employment types.
  5. "Economic Impact": Broader macroeconomic motivations, such as increasing employment, encouraging investment, or reducing public debt ratios.
  6. "Inflation or Increase in Prices": Reforms responding to the erosion of real pension values due to sustained or unexpected price increases.

7. "Demographic Changes": Reforms motivated by trends such as aging populations, declining fertility, or increasing life expectancy.
8. "Inequality and Fairness": Reforms targeting unequal treatment or disparities in pension outcomes between population groups.
9. "Labor Market Changes": Reforms introduced in response to changes in employment patterns, such as informalization, gig work, or declining participation rates.
10. "Other": Residual category for reform motivations not captured by the above, or where the motivation is ambiguous or mixed.

- **Retirement Age Change Details**
  Details the specific nature and magnitude of the statutory retirement age change. This includes which groups (e.g., male/female, public/private sector workers) were affected and the size of the increase or decrease in retirement age. Phased adjustments or future-scheduled age shifts are also noted.
- **Retirement Age Pre Reform Female**
  The statutory retirement age for women prior to the implementation of the reform.
- **Retirement Age After Reform Female**
  The statutory retirement age for women following the implementation of the reform.
- **Retirement Age Pre Reform Male**
  The statutory retirement age for men prior to the implementation of the reform.
- **Retirement Age After Reform Male**
  The statutory retirement age for men following the implementation of the reform.
- **Contribution Rate Change**
  Indicates whether there was a change in contribution rates—either an increase, decrease, or no change—for employers, employees, or both. This variable may also include qualitative details about how the rate changed and which groups were affected.
- **Old Employer Contribution Rate**
  The employer's statutory contribution rate (as a percentage of covered wages) before the reform took effect.
- **New Employer Contribution Rate**
  The employer's statutory contribution rate after the reform took effect.
- **Old Employee Contribution Rate**
  The employee's statutory contribution rate before the reform took effect.
- **New Employee Contribution Rate**
  The employee's statutory contribution rate after the reform took effect.
- **Old Contribution Rate (Employee and Employer Combined)**
  The total statutory pension contribution rate—combining both employer and employee shares—prior to the reform.
- **New Contribution Rate (Employee and Employer Combined)**
  The total statutory pension contribution rate—combining both employer and employee shares—after the reform.

B. A Bird's-Eye View

To provide a bird's-eye view of the GPRD, this section illustrates the global evolution of pension reform activity by constructing a series of world maps (Figure 1-6) in which reform frequency is visualized using proportional bubbles overlaid on countries. Each map corresponds to a benchmark year: 1970, 1980, 1990, 2000, 2010,

and 2025, and captures the cumulative number of pension reform episodes (measures) recorded up to that point. The size of each bubble reflects the total number of announced, legislated, implemented, or reversed pension reform measures in a given country by the respective cutoff year. This visual representation highlights both the geographic diffusion and the temporal intensification of reform efforts. In earlier decades, reform activity was primarily concentrated in Western Europe and parts of Latin America. Over time, reform momentum expanded to emerging and developing economies, with notable increases in Eastern Europe, Sub-Saharan Africa, and parts of Asia after the 1990s. By 2025, nearly all regions exhibit active reform histories, with the largest bubbles concentrated in countries such as Argentina, Brazil, China, and France, each of which has experienced sustained and recurrent policy interventions. These maps offer a visually compelling depiction of pension reform as a truly global policy phenomenon, shaped by evolving fiscal, demographic, and institutional pressures.

Figure 1. Evolution of Global Pension Reforms as of April 1970



Source: GPRD.

Note: Bubble size indicates the frequency of pension reform episodes (announcement, legislation, implementation, and reversal).

Figure 1 shows that by 1970, pension reform activity was largely concentrated in advanced economies (identified as advanced using the latest, April 2023, World Economic Outlook classification). Western European countries such as France, Germany, and the United Kingdom were early adopters of pension legislation adjustments, often focused on expanding coverage and introducing earnings-related schemes, predominantly defined-benefit (DB) arrangements that linked benefits to workers' earnings histories. Latin American countries like Argentina and Brazil also showed early reform activity, driven by industrialization and expanding formal labor markets. Reform episodes during this period were primarily foundational, aimed at building or consolidating national systems.

Figure 2. Evolution of Global Pension Reforms as of April 1980

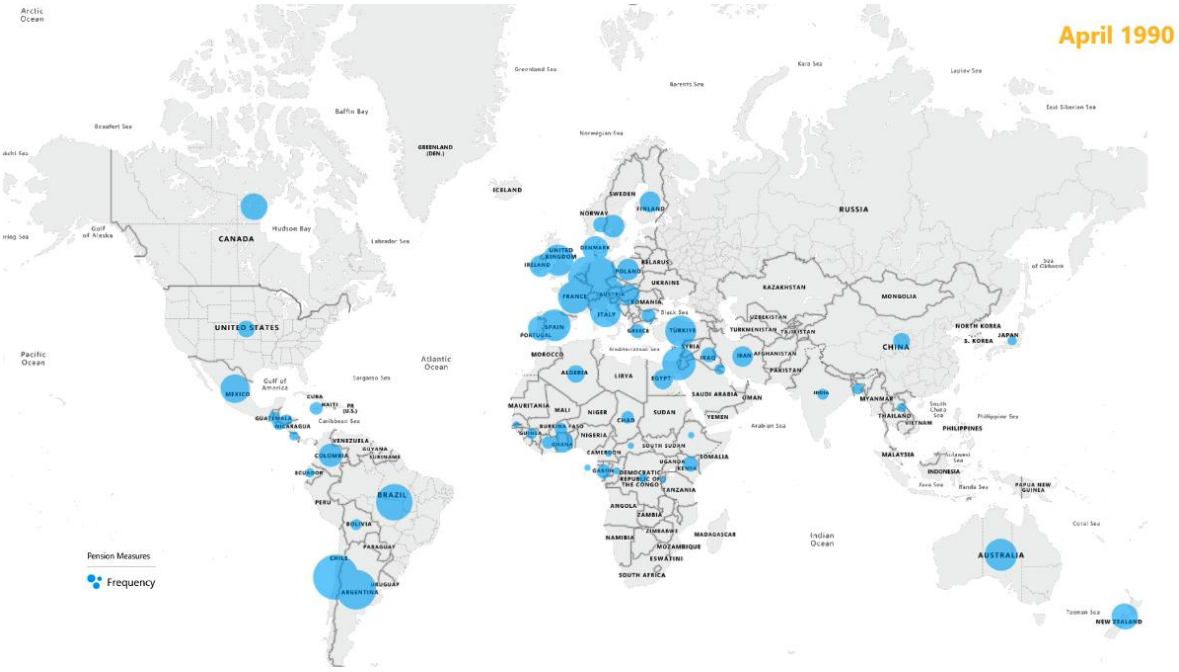


Source: GPRD.

Note: Bubble size indicates the frequency of pension reform episodes (announcement, legislation, implementation, and reversal).

The 1980 map above (Figure 2) reflects a modest expansion in reform geography. While Western Europe and Latin America continued to account for most activity, new reform episodes emerged in countries such as Australia, Japan, and select Eastern European states. The 1970s oil shocks and global inflation prompted many governments to revise benefit indexation mechanisms, while some began to address early concerns around system sustainability. Nevertheless, reform intensity remained relatively low in Africa and Asia.

Figure 3. Evolution of Global Pension Reforms as of April 1990



Source: GPRD.

Note: Bubble size indicates the frequency of pension reform episodes (announcement, legislation, implementation, and reversal).

By 1990 (Figure 3 above), a new wave of reforms had emerged, particularly in Latin America and Eastern Europe. Countries like Chile had initiated path-breaking structural reforms in the early 1980s, transitioning to funded defined contribution systems. These reforms inspired similar initiatives across the region. In Eastern Europe, the dissolution of socialist regimes prompted initial restructuring of pay-as-you-go (PAYG) systems. Reform diffusion was still uneven, with limited activity in Sub-Saharan Africa and developing Asia.

Figure 4. Evolution of Global Pension Reforms as of April 2000

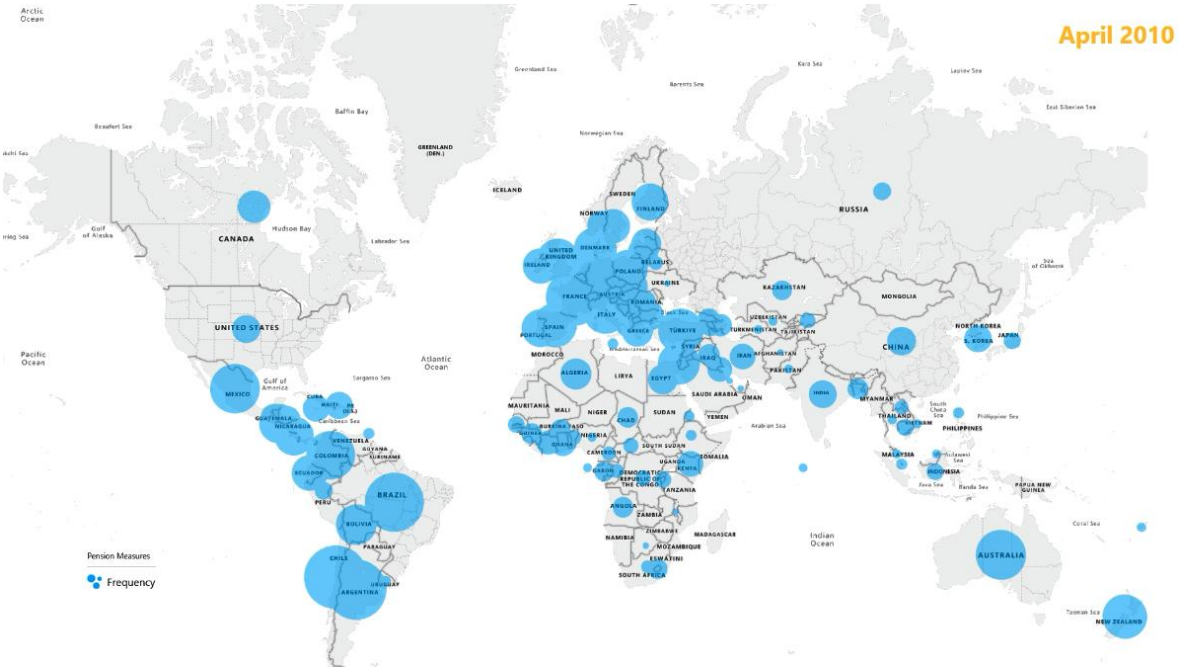


Source: GPRD.

Note: Bubble size indicates the frequency of pension reform episodes (announcement, legislation, implementation, and reversal).

The year 2000 (Figure 4) marks a significant increase in global pension reform activity. Latin American and Eastern European countries, including Poland, Hungary, and Kazakhstan, intensified structural transitions toward multi-pillar systems. Reforms began to spread more broadly across Asia and Africa, often supported by international financial institutions. Key reform types during this period included raising retirement ages, modifying benefit formulas, and introducing mandatory savings components. Pension reform became a mainstream macro-fiscal concern in both emerging and advanced economies.

Figure 5. Evolution of Global Pension Reforms as of April 2010

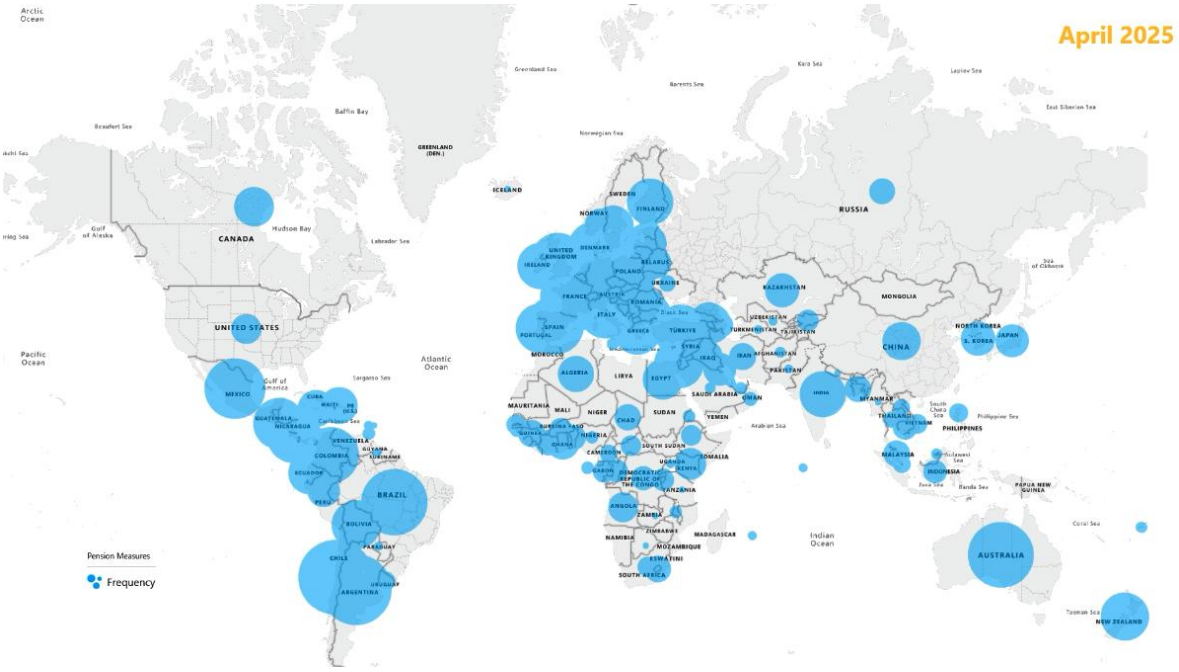


Source: GPRD.

Note: Bubble size indicates the frequency of pension reform episodes (announcement, legislation, implementation, and reversal).

By 2010 (Figure 5), reform activity had become increasingly global in scope. The 2008 global financial crisis triggered widespread fiscal consolidation measures, prompting reforms in countries like Greece, Spain, and Portugal. Advanced economies began focusing on parametric adjustments, raising retirement ages, tightening eligibility, and altering indexation rules, while emerging economies continued structural reforms to expand coverage and improve adequacy. Notably, pension portability, inclusion of the informal sector, and gender equity became growing priorities in the reform agenda.

Figure 6. Evolution of Global Pension Reforms as of April 2025



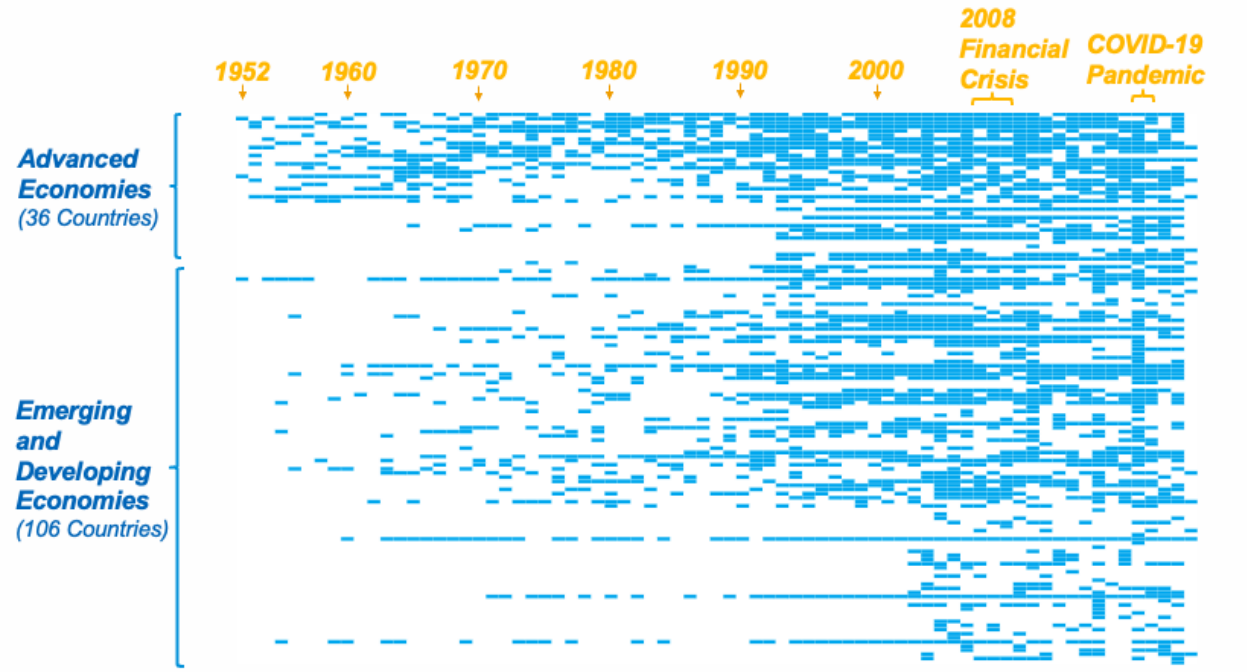
Source: GPRD.

Note: Bubble size indicates the frequency of pension reform episodes (announcement, legislation, implementation, and reversal).

The 2025 map (Figure 6) captures a steep rise in both the frequency and geographic breadth of pension reforms. Countries across all income groups exhibit sustained reform activity. In addition to parametric and structural changes, newer themes such as digital administration, automatic stabilizers, and climate-resilient social protection systems have begun to appear. Major reform waves are visible in populous countries like China, India, and Brazil. Pension reform has now evolved into a dynamic and iterative policy process shaped by fiscal stress, demographic aging, labor market shifts, and global uncertainty.

To complement the spatial patterns captured in the world maps, a reform timeline chart was constructed (Figure 7) that visualizes the annual distribution of pension reform activity across 36 advanced economies and 106 emerging and developing economies from 1952 to 2025, following the IMF's World Economic Outlook classification. Each row represents a country, while each column indicates a calendar year; blue cells mark years in which at least one reform episode was recorded, and white cells indicate years without reform activity. This matrix-style visualization reveals how pension reform has evolved as a recurring policy intervention over time. Advanced economies display earlier and more consistent reform engagement, with concentrated activity beginning in the postwar decades and extending through multiple fiscal cycles. In contrast, pension reform among emerging and developing economies remains sparse until the early 1990s, after which activity accelerates, often in response to externally anchored programs or domestic institution-building. Temporal clustering is particularly pronounced around major global crises. For example, the 2008 global financial crisis coincides with a dense wave of reforms, especially in Europe and Latin America, while the 2020 COVID-19 shock triggered widespread pension adjustments across both advanced and emerging economies. The chart underscores not only the growing universality of pension reform as a policy tool but also its cyclical responsiveness to external shocks and macro-fiscal pressures.

Figure 7. Global Pension Reform Timeline from 1952 to 2025



Source: GPRD.

C. Comparison

To evaluate the reliability of the Global Pension Reform Database (GPRD) and position it relative to existing sources, it was compared with two prominent datasets in the pension reform literature (Table 1). The first, compiled by Bi and Zubairy (2023), records changes in public pension policy for ten OECD countries over 1961–2019. Their dataset was constructed manually, drawing primarily on OECD *Economic Surveys* and legislative archives, and codes each reform episode along four dimensions: the direction of generosity change, the policy instruments employed (such as benefit formulas, coverage, indexation rules, and retirement age), the stated policy rationale, and the lag between enactment and full implementation.

The second dataset is Romp (2024), which integrates and extends the work of Bi and Zubairy (2023) and Beetsma et al. (2020). Romp broadens coverage to 23 OECD countries over 1970–2023, incorporates both fiscal and participant-oriented reforms, adds reforms affecting occupational pension schemes, and extends the temporal scope beyond 2017. For the ten countries common to both datasets, the recorded reform years align in roughly 70 percent of cases between 1970 and 2019, with most of the reforms in Bi and Zubairy also appearing in Romp, while the latter captures additional episodes due to its broader definitions and data sources (Table 2).

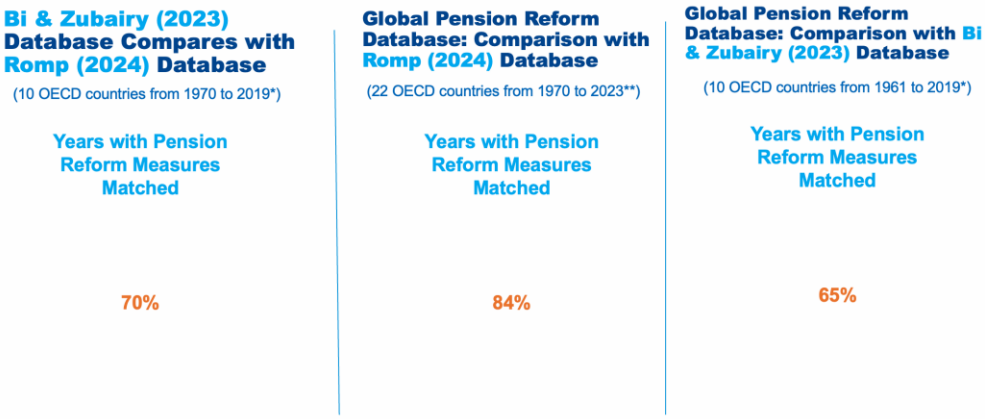
When benchmarked against these datasets, the GPRD matches 65 percent of reform years in Bi and Zubairy (2023) for the ten overlapping countries over 1961–2019, and 84 percent of reform years in Romp (2024) for the 23 overlapping countries over 1970–2023. These levels of agreement underscore the robustness of the GPRD's large-scale extraction and coding approach, while remaining discrepancies reflect differences in scope, definitional choices, and underlying source material.

Table 1. Comparison of the GPRD with Existing Pension Reform Datasets

Feature	Bi & Zubairy (2023)	Romp (2024)	Global Pension Reform Database (2025)
Coverage	10 OECD countries (1961–2019)	23 OECD countries (1970–2023)	**142 countries** (1952–2025)
Collection Method	Manual extraction from OECD Economic Surveys and national laws	Manual extraction from ILO NATLEX, ISSA, OECD Economic Surveys, and EC LABREF	**Ring-fenced LLM-based extraction** of 25,000 EIU country reports and 260 SSA reports + human Quality Assurance
Granularity	Individual pension reform episodes; 4 fields (sign, tool, motive, lag)	Individual pension reform packages: generosity sign; phase-in length	Individual reform measures; **~21 variables** (status, type, contribution-rate change, retirement-age change)
Update Frequency	One-off (2019)	One-off (2023)	Quarterly push-button updates (zero marginal cost)
Automation & Scale	Fully manual	Semi-auto + manual	**Automated pipeline**

Source: Bi and Zubairy (2023), Romp (2024) and authors' calculations.

Table 2. Alignment of Reform Years in the GPRD with Existing Datasets



*The 10 countries are Australia, Belgium, Denmark, Finland, France, Italy, Japan, New Zealand, Spain, and the United Kingdom.

**The 22 countries are: Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, and the United States. Luxembourg is the only country that is in Romp (2024) but not in our GPRD.

Source: Bi and Zubairy (2023), Romp (2024) and authors' calculations.

D. Caveats and Database Limitations

While the GPRD offers broad coverage across countries and time, several important limitations remain. The database relies primarily on EIU and SSA narrative reports, and the coverage and level of detail vary across countries and over time. Technical changes and developments in the private pillar are often underreported, so some reforms, particularly those relating to Pillar II, tax and incentive design, or consumer protection, may be recorded with lower confidence. Our prompt-based LLM extraction pipeline enables large-scale coding but can misclassify reform status, timing, or type, hallucinate events, or miss context-specific details. Despite extensive manual review and cleaning, residual measurement error is inevitable given the dataset's scope, which includes over 4,000 reform entries and 21 reform-related variables across 142 countries. For example, pension indexation is a routine, legislated practice in many countries but may be reported in source narratives only when unusually large, for example during periods of high inflation. Such reports do not necessarily constitute a reform and can lead to overstated reform counts.

V. Stylized Facts

A. Global Rise and Convergence in Pension Reform Engagement

Pension reform has shifted from a policy instrument confined to advanced economies into a globally recurring response to fiscal and demographic pressure. As shown in Figure 8, between 1952 and 1989, the average annual share of countries undertaking a pension reform was 19.5 percent in advanced economies, 7.3 percent in emerging markets, and 3.3 percent in developing economies. This pattern changed significantly after 1990, with the averages rising to 46.0, 33.2, and 18.5 percent, respectively, over the 1990 to 2012 period. The most active reform phase occurred between 2003 and 2012, reaching 59.8 percent in advanced economies, 47.0 percent in emerging markets, and 27.8 percent in developing economies.

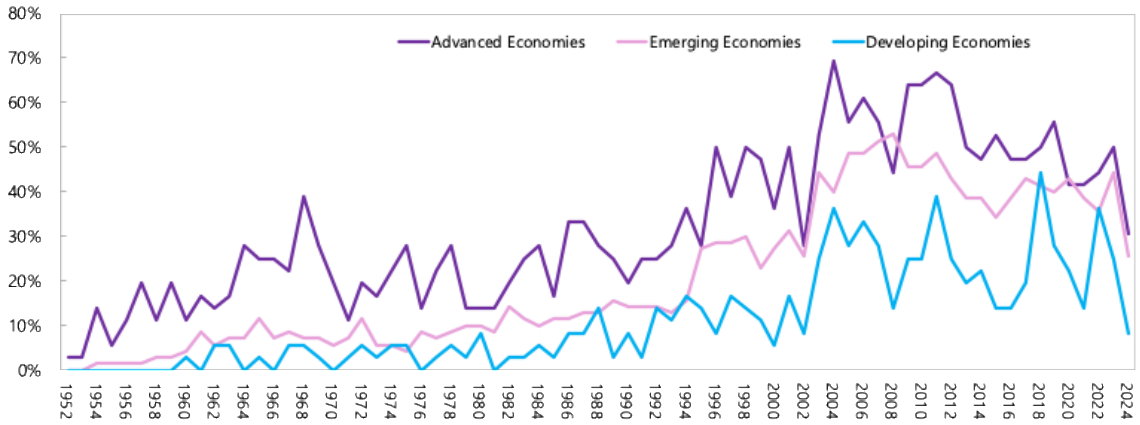
Although the timing of reform peaks varied across groups, the pattern reflects convergence around global crisis cycles. Advanced economies peaked in 2004, ahead of the 2008 global financial crisis, which triggered a sharp rise in reform among emerging markets, peaking at 53 percent in 2008. Developing economies reached their highest reform share in 2018, following a wave of debt sustainability concerns and fiscal adjustment programs. More recently, the COVID-19 pandemic placed unprecedented strain on social protection systems and public finances, prompting temporary pension measures in both high- and middle-income countries. Meanwhile, debt distress in several emerging and developing economies, such as Argentina, Kenya, and Sri Lanka, has revived attention to pension reform as part of broader fiscal consolidation packages.

Year-to-year volatility in reform participation increases with lower income levels. The coefficient of variation is 0.53 in advanced economies, 0.79 in emerging markets, and 0.98 in developing economies, reflecting more episodic and reactive reform patterns where institutional capacity and political constraints limit sustained engagement. The reform gap between advanced and developing economies widened after 1990, growing from 16 percentage points before 1990 to 28 percentage points between 1990 and 2012.

Since 2013, reform momentum has moderated. From 2020 to 2024, the average share of countries implementing reform declined to 41.8 percent in advanced economies, 37.6 percent in emerging markets, and 21.0 percent in developing economies. However, this is still well above levels seen in the 1990s.[4] The figure captures both the global diffusion of pension reform and the differentiated ability of countries to sustain reform cycles, shaped by the depth of institutions, fiscal space, and responsiveness to macroeconomic shocks.

[4] It should be noted that the persistently elevated level can also be due to more frequent reporting on pension reforms that would previously have flown under the media radar.

Figure 8. Share of Countries with Pension Reform Episodes, by Income Group, 1952-2024



Sources: Global Pension Reform database; and authors' calculations.
Note: The figure presents the average annual share of countries implementing pension reform measures within each income group for a sample of 142 countries. Advanced economies are classified according to the IMF World Economic Outlook (WEO) as of April 2025, developing economies are IMF PRGT-eligible countries as of March 2025, and emerging economies comprise all remaining countries.

B. Reform Timelines and Labor Market Implications

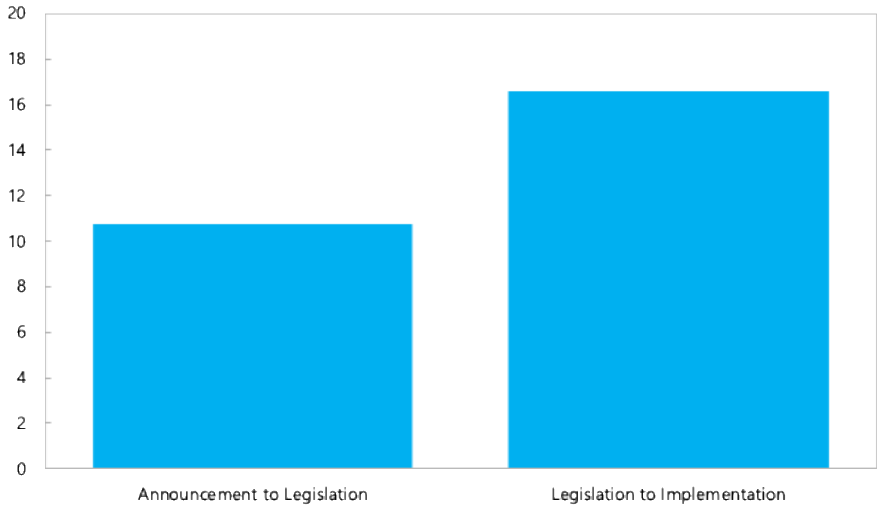
Pension reform unfolds through a multi-stage process, typically extending over several years from initial announcement to full implementation. As shown in Figure 9, the average time between announcement and legislative approval is approximately 11 months, while the transition from legislation to implementation takes around 17 months. These delays reflect not only administrative and legal bottlenecks but also political economy frictions such as coalition-building, stakeholder resistance, and public consultation. The cumulative lag of nearly 28 months highlights the medium-run nature of pension reform cycles and the limits of using such reforms as short-run stabilization tools.[5]

The extended implementation window also has direct implications for labor market behavior. When reforms involve increases in retirement age, tighter eligibility rules, or changes in contribution rates, the anticipation of future changes may influence current labor supply decisions. Workers nearing retirement may adjust their exit timing to avoid less generous benefit terms, while younger cohorts may alter savings behavior or labor force participation based on perceived returns to formal employment. For example, delayed implementation of stricter retirement conditions can lead to short-term retirement surges, undermining intended fiscal gains. Conversely, long lead times may improve outcomes by giving firms and workers time to adapt to new rules, especially in settings where pension coverage is incomplete or labor markets are segmented.

The duration from legislation to implementation is particularly critical in developing and emerging economies, where administrative readiness and labor informality complicate enforcement. Implementation delays in these contexts may dampen reform credibility, weaken compliance, or reinforce dual labor market structures. Understanding these timing patterns is therefore essential not only for forecasting fiscal impacts, but also for anticipating short-term distortions and long-term behavioral responses in labor markets.

[5] In addition to the lengthy and complex legislative process, pension reforms typically have only a minor short-term impact on economic outcomes such as public expenditure because acquired rights are difficult to modify. As a result, reforms often depend on attrition effects to achieve savings over the medium and long term

Figure 9. Average Duration (in Months) Between Pension Reform Stages: Announcement, Legislation, and Implementation



Sources: Global Pension Reform database; and authors' calculations.
Notes: Average durations are calculated based on pension reform episodes from 142 countries over the period 1952–2025. Each reform episode is clearly identified by a unique reform ID. Durations are computed as the number of months between the dates of announcement and legislation, and separately between legislation and implementation, and then averaged across all valid observations. Reform episodes lacking subsequent stages or containing negative durations (due to data inconsistencies) were excluded from the calculation.

C. Shifts in the Generosity of Legislated Pension Reforms

Pension reform design by policymakers faces a tradeoff between fiscal sustainability and benefit adequacy. Figure 10 presents the annual share of reforms classified as either less generous or more generous, based on observed changes to replacement rates, eligibility criteria, and benefit formulas. In the early decades, particularly from the 1950s through the mid-1970s, reforms were predominantly expansive, with the majority of recorded episodes enhancing benefit levels or easing access. This pattern corresponds to the postwar expansion of contributory systems and the consolidation of welfare state institutions, primarily in advanced economies.

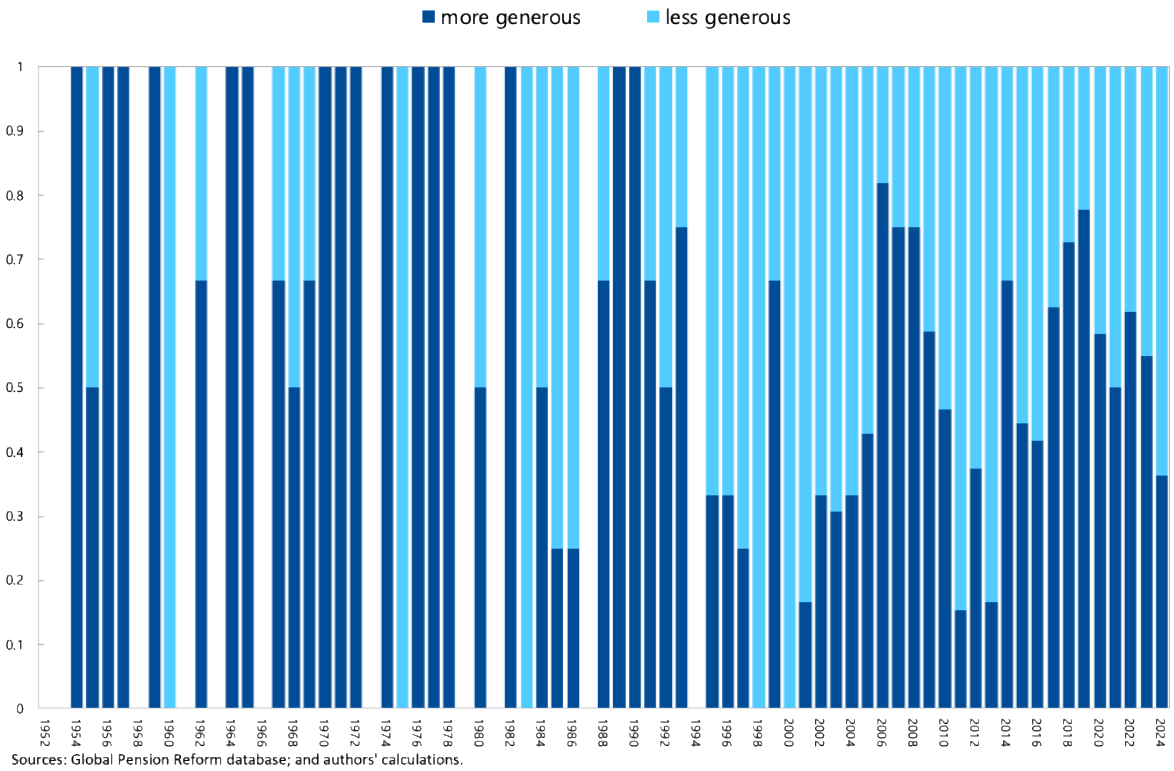
A shift becomes evident in the late 1980s and early 1990s, coinciding with the onset of population aging, growing pension liabilities, and rising debt burdens in many countries. Between 1995 and 2005, less generous reforms accounted for a majority of identified cases in most years, reaching over 80 percent in 2001. Reforms during this period frequently involved tightening eligibility, modifying indexation rules, or reducing accrual rates, often under the banner of ensuring long-term sustainability. The global financial crisis further reinforced this trend, as fiscal consolidation became a dominant policy objective in both advanced and emerging markets.

Since 2010, reform direction has been more heterogeneous. While retrenchment remains prominent in several high-income countries, a growing number of emerging and developing economies have adopted more generous measures, particularly during and after the COVID-19 crisis. For instance, in 2018 and 2019, over 70 percent of reforms expanded benefits, often as part of efforts to strengthen social protection or extend coverage. In contrast, the most recent data for 2024 show a reversion toward retrenchment, with 64 percent of reforms classified as less generous.

These shifts carry important implications for labor market behavior. Retrenchments may increase labor force participation among older workers or shift retirement timing, while expansions, particularly when targeted, can strengthen formalization incentives and improve income security. The cyclical pattern observed in reform

generosity suggests that pension policy remains closely linked to broader fiscal and macroeconomic dynamics. While the classification relies on textual analysis and may understate marginal changes, the directional trends are consistent with structural transformations in pension systems worldwide.

Figure 10. Share of Legislated Pension Reforms with Benefits Classified as Less vs. More Generous, 1952–2024



Sources: Global Pension Reform database; and authors' calculations.
Notes: The shares are calculated annually from legislated pension reform episodes for 142 countries over the period 1952–2024. Each reform was classified as either "less generous" or "more generous" based on the "benefit_change" variable, which captures adjustments to replacement rates, eligibility criteria, and benefit formulas. The annual share represents the proportion of reforms in each generosity category relative to the total number of legislated reforms with clearly identified benefit changes. Due to the nature of the data extraction process involving Large Language Models (LLM), the "benefit_change" classification may not be comprehensive.

VI. Empirical Analysis

This section illustrates how the Global Pension Reform Database can be used to study the macro-fiscal consequences of pension reform. The empirical results are presented as an illustrative application of the GPRD rather than a comprehensive causal evaluation. The focus is on the evolution of public debt following the implementation of pension reforms, with particular attention to differences between advanced and non-advanced economies. Beyond the fiscal example shown here, the GPRD can support many further research

questions, including the growth implications of pension reform and effects on labor-market outcomes, but also the drivers behind pension reforms.[6] These extensions are left for future work using the database.

The empirical analysis addresses four key questions: (i) Do pension reforms have a measurable effect on public debt ratios? (ii) Are these effects heterogeneous across income groups? (iii) What is the dynamic pattern of debt adjustment around reform episodes? and (iv) Are the results robust to concerns about pre-existing fiscal conditions and debt persistence?

To answer these questions, a difference-in-differences (DiD) design is employed using a balanced panel of 142 countries from 1990 to 2024, supplemented by an event-study framework to examine dynamic effects. The identifying variation comes from the timing of the most recent implemented pension reform in each country, as recorded in the database. The analysis is intended as a stylized empirical application rather than a causal evaluation of specific reform types. It demonstrates how the GPRD can be leveraged to support empirical work on structural reforms in a macroeconomic context.

Table 3 reports summary statistics for the analysis sample, separately for advanced and non-advanced economies. Advanced economies carry higher average debt (64.8 versus 51.8 percent of GDP) and reform more frequently (32.5 versus 18.8 percent of country-years), while non-advanced economies exhibit higher macroeconomic volatility and substantially greater IMF program exposure (37.6 versus 6.3 percent of observations).

Table 3. Summary Statistics (1990–2024)

	Advanced	Non-Advanced	Full Sample
Panel A: Sample Coverage			
Countries	36	106	142
Country-years	1260	3710	4970
Obs. with Debt/GDP data	1043	2647	3690
Panel B: Debt/GDP (% of GDP)			
Mean	64.8	51.8	55.5
Median	57.4	43.8	46.8
Std. Dev.	41.0	35.0	37.3
Min	3.8	1.5	1.5
Max	258.4	357.7	357.7
Panel C: Year-on-Year Change in Debt/GDP (p.p.)			
Mean	0.89	-0.05	0.22
Median	-0.01	0.01	0.01
Std. Dev.	6.07	12.32	10.93
Panel D: Pension Reform Frequency			

[6] Past research on pension reforms has examined both the drivers (demographic, economic, fiscal) and their effects on economic and fiscal outcomes, including labor market participation, savings behavior, economic growth, public debt, and pension system sustainability.

Implemented reforms (count)	410	698	1108
Reform frequency (%)	32.5	18.8	22.3
Parametric reforms (count)	263	315	578
Structural reforms (count)	110	185	295
Panel E: Control Variables (means)			
GDP growth (%)	2.62	3.74	3.46
GDP per capita growth (%)	1.79	2.08	2.00
Inflation, CPI (%)	8.3	37.5	29.5
Old-age dependency ratio	23.1	9.3	12.9
Unemployment rate (%)	7.4	7.8	7.7
IMF program (% of obs.)	6.3	37.6	29.7
Gini (disposable income)	29.7	41.1	37.6

Notes: Reform frequency is the share of country-years with at least one implemented pension reform. Control variable means are computed over all non-missing observations in each group.

A. Baseline DiD Results

As a first step, the average effect of the last implemented pension reform is estimated using the following specification:

$$\mathrm{Debt/GDP}_{it} = \beta \cdot post_{it} + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

where $post_{it}$ is an indicator equal to 1 if country i is at or beyond the year of its most recent implemented pension reform (this approach indexes to the most recent reform to avoid post-treatment contamination from subsequent reforms and to focus on the most contemporaneous, policy-relevant change), X_{it} includes time-varying controls (GDP growth, GDP per capita growth, inflation, the old-age dependency ratio, and a lagged IMF program dummy), $\mathrm{Debt/GDP}_{it}$ is government gross debt (percent of GDP) for each country-year observation. α_i and λ_t are country and year fixed effects. Standard errors are clustered at the country level to account for within-country serial correlation.[7]

The results, reported in Table 4, reveal a marked difference between advanced and non-advanced economies. In advanced countries, the coefficient on $\mathrm{post}_{\mathrm{it}}$ is statistically insignificant (β = 0.079, SE = 1.671), indicating no discernible change in debt levels following reforms. By contrast, non-advanced economies exhibit a statistically significant reduction in debt-to-GDP ratios following implemented pension reforms (β = –2.960, SE = 1.387, p < 0.05).[8] These findings suggest that pension reforms may be more effective in curbing debt accumulation in non-advanced economies, potentially reflecting stronger fiscal discipline or external conditionalities in those contexts.

[7] The independent variables do not explicitly correct for potential simultaneous structural reforms on the labor, product or capital market or tax reforms. However, some of these will be picked up by the IMF program dummy.

[8] This is despite the country and year fixed effects absorbing a lot of the variation

B. Event Study Analysis

To examine the dynamic effects of pension reforms and assess the validity of the parallel trends assumption, this paper estimates an event-study specification using relative time indicators covering the period from five years before to five years after the last implemented pension reform. The year prior to reform (k = -1) is omitted and serves as the reference category. The specification includes country and year fixed effects, as well as the same set of controls used in the baseline difference-in-differences model.

The event-study plot (Figure 11) for non-advanced countries reveals two main insights:

1. Pre-trends Test: A joint test of the coefficients on the four pre-treatment periods (k = -5 to k = -2) yields an F-statistic of 0.84 (p = 0.506), indicating no statistically significant pre-trend, which supports the validity of the identifying assumption of parallel trends.
2. Dynamic Effects: The average post-treatment effect across years k = 0 to k = +5 is significantly negative at –3.94 percentage points (SE = 1.49, p ≈ 0.01), confirming a gradual and sustained decline in the debt-to-GDP ratio following reform implementation.

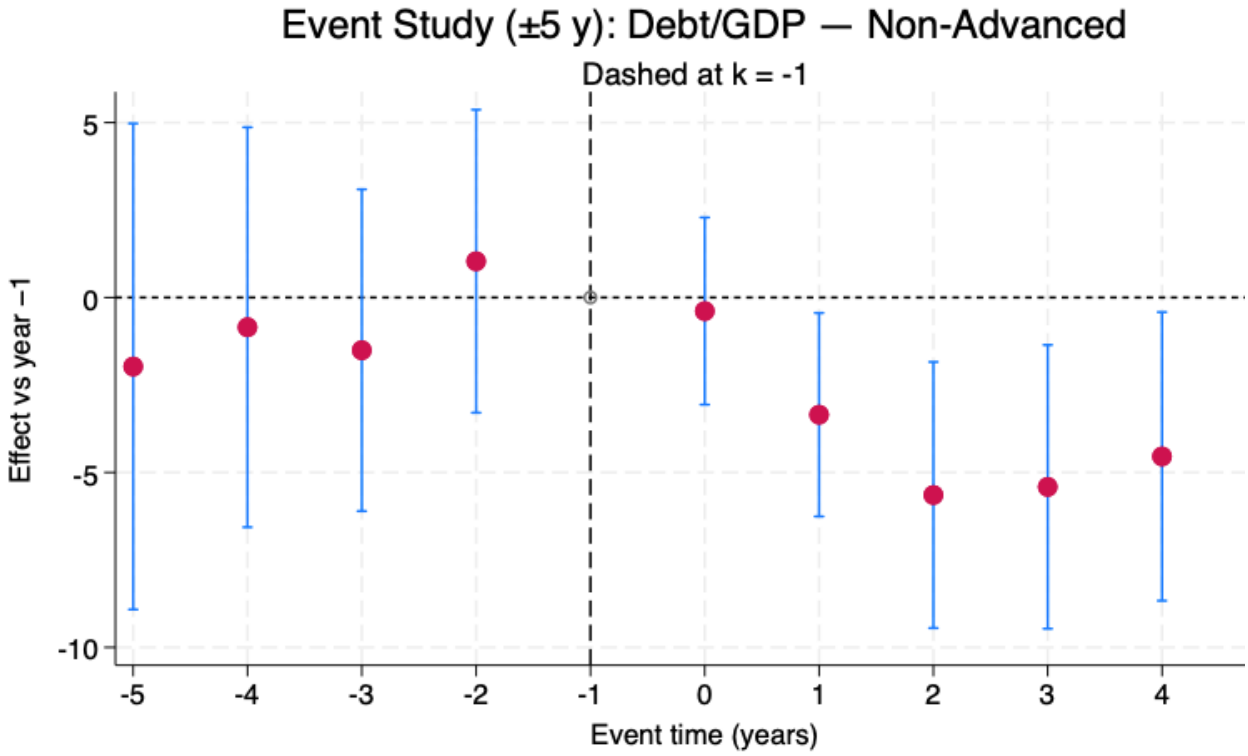
These results provide evidence that in non-advanced countries, pension reforms are followed by statistically and economically meaningful reductions in public debt. The corresponding event-study analysis for advanced economies, presented in Annex Figure A1, confirms the null result: all coefficients are statistically insignificant and the average post-reform effect is close to zero (−0.86 p.p., p = 0.717).

Table 4. TWFE DiD Estimates: Effect of Pension Reform on Debt-to-GDP (±5 Years)

	(1) Advanced Economies	(2) Non-Advanced Economies
	Debt/GDP	Debt/GDP
Post-Last-Reform	0.079	-2.960**
	(1.671)	(1.387)
Observations	189	487
R-squared	0.993	0.909
Country FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
Clustered SEs	By country	By country

Notes: Controls include GDP growth, inflation, old-age dependency ratio, lagged IMF program dummy, and GDP per capita growth. GDP growth, inflation, GDP per capita, and IMF program dummy are sourced from the IMF. The old-age dependency ratio is from the World Bank. Standard errors are in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01

Figure 11. Event Study for Non-Advanced Economies



Notes: The figure plots event-study coefficients from a TWFE regression of debt-to-GDP on relative-time indicators around the last implemented pension reform for non-advanced economies. The omitted category is k = −1. Vertical bars represent 95 percent confidence intervals. Standard errors are clustered at the country level. Controls include GDP growth, GDP per capita growth, inflation, the old-age dependency ratio, and a lagged IMF program dummy.

C. Robustness Checks

To evaluate the robustness of the baseline DiD estimates, a series of exercises addressing two main threats to identification is conducted: (i) confounding from differences in fiscal space, and (ii) dynamic persistence in debt series. Table 5 presents results from three alternative specifications. Specification A augments the baseline controls with the primary fiscal balance (percent of GDP) to account for pre-existing differences in fiscal space. Specification B adds a one-year lagged debt-to-GDP ratio to capture serial correlation and short-run debt adjustments. Specification C includes both additional controls.

The substantive conclusion is stable across these specifications. In the non-advanced sample the estimated post-reform decline in debt-to-GDP remains economically meaningful and statistically significant: the treatment coefficient is approximately −2.9 percentage points in Specifications A–C (Table 5, cols. 2, 4 and 6) and is significant at conventional levels. By contrast, the corresponding coefficients for advanced economies are small and statistically indistinguishable from zero in all specifications (Table 5, cols. 1, 3 and 5). The stability of estimates across Specifications A–C indicates that neither initial fiscal position nor short-run debt persistence materially changes the inference that pension reforms are followed by debt reductions in non-advanced economies, while no comparable effect is evident for advanced economies.

Table 5. Robustness Checks: Effect of Last Pension Reform on Debt-to-GDP (±5 years)

	(1) Advanced (A)	(2) Non-Advanced (A)	(3) Advanced (B)	(4) Non-Advanced (B)	(5) Advanced (C)	(6) Non-Advanced (C)
	Debt/GDP	Debt/GDP	Debt/GDP	Debt/GDP	Debt/GDP	Debt/GDP
Post-Last-Reform	0.870	-2.941**	-0.063	-2.930**	0.386	-2.904**
	(1.577)	(1.363)	(1.099)	(1.360)	(1.041)	(1.333)
Observations	189	487	189	487	189	487
R-squared	0.994	0.909	0.997	0.912	0.997	0.913
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Primary Balance	✓	✓			✓	✓
Lagged Debt/GDP			✓	✓	✓	✓
Clustered SEs	By country	By country	By country	By country	By country	By country

Notes: All regressions include country and year fixed effects. Standard errors in parentheses are clustered at the country level. "Primary Balance" refers to primary_balance_gdp; "Lagged Debt/GDP" refers to L1_debt_gdp. Controls in all columns: GDP growth, inflation (CPI), old-age dependency ratio, lagged IMF program dummy, and GDP per capita growth. GDP growth, inflation, GDP per capita, the primary fiscal balance, and IMF program dummy are sourced from the IMF. The old-age dependency ratio is from the World Bank. Significance levels: * p < 0.10, ** p < 0.05, *** p < 0.01.

VII. Conclusion

The Global Pension Reform Database (GPRD) provides the first systematic record of pension reforms across advanced, emerging, and developing economies over more than seven decades. By extracting and classifying over 4,000 reform episodes from *Economist Intelligence Unit* country reports and *Social Security Administration International Updates* using a ring-fenced large language model, the GPRD extends coverage well beyond OECD economies and offers a consistent basis for comparative analysis across countries and over time.

The evidence from the GPRD reveals that pension reform has evolved into a global and recurring policy tool. Reform engagement has broadened markedly since the 1990s, with activity increasingly aligned with macroeconomic shocks, fiscal pressures, and demographic change. The direction of reform has shifted from postwar expansion to periods of retrenchment, followed more recently by a mix of both approaches.

The empirical application illustrates the database's utility for macro-fiscal analysis. In non-advanced economies, implemented pension reforms are followed by statistically significant reductions in public debt, while no comparable effects are detected for advanced economies. This divergence underscores the importance of initial fiscal conditions, institutional frameworks, and implementation capacity in shaping outcomes.

The GPRD is subject to certain limitations. These include possible sampling noise from the frequency of EIU reports, underreporting of minor reforms, overidentification from reporting of non-reform pension events, and variability in narrative detail across sources. Although extensive quality checks were conducted, some errors are inevitable given the scope of the dataset. Future work should incorporate additional sources, improve coding precision, and further validate coverage.

By consolidating a global record of pension reforms, the GPRD offers a foundation for evidence-based policy design. Expanding and refining the database will strengthen its value for assessing reform timing, design, and macroeconomic impacts, supporting the development of pension systems that are fiscally sustainable and responsive to changing demographic and economic conditions.

Annex I. Countries Included in the Global Pension Reform Database

List of Countries Included in the Global Pension Reform Database (GPRD)

Afghanistan
Albania
Algeria
Andorra
Angola
Argentina
Armenia
Australia
Austria
Azerbaijan
Bahrain
Bangladesh
Barbados
Belarus
Belgium
Belize
Benin
Bolivia
Bosnia and Herzegovina
Botswana
Brazil
Brunei Darussalam
Bulgaria
Burkina Faso
Burundi
Cambodia
Cameroon
Canada
Central African Republic
Chad
Chile
China
Colombia
Congo, Republic of
Israel
Italy
Jamaica
Japan
Jordan
Kazakhstan
Kenya
Kosovo
Kuwait
Kyrgyz Republic
Lao P.D.R.
Latvia
Lebanon
Lesotho
Liechtenstein
Lithuania
Luxembourg
Malawi
Malaysia
Maldives
Malta
Mauritius
Mexico
Moldova
Monaco
Myanmar
Nepal
Netherlands
New Zealand
Nicaragua
Nigeria
North Macedonia
Norway
Oman

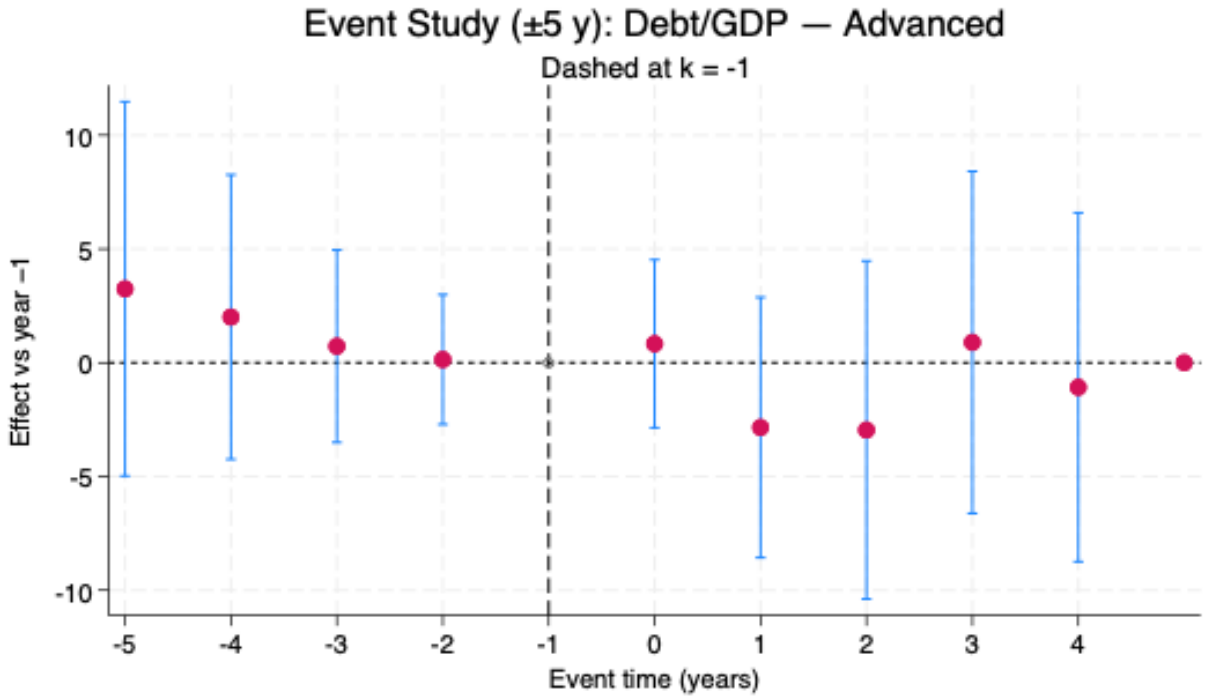
Costa Rica	Pakistan
Côte d'Ivoire	Panama
Croatia	Paraguay
Cuba	Peru
Cyprus	Philippines
Czech Republic	Poland
Democratic Republic of the Congo	Portugal
Denmark	Qatar
Dominican Republic	Romania
Ecuador	Russia
Egypt	São Tomé and Príncipe
El Salvador	Saudi Arabia
Eritrea	Serbia
Estonia	Singapore
Ethiopia	Slovak Republic
Fiji	Slovenia
Finland	South Africa
France	South Korea
Gabon	Spain
Gambia, The	St. Vincent and the Grenadines
Georgia	Sweden
Germany	Switzerland
Ghana	Tanzania
Greece	Thailand
Guatemala	Trinidad and Tobago
Guinea	Türkiye
Guinea-Bissau	Turkmenistan
Guyana	Uganda
Haiti	Ukraine
Honduras	United Arab Emirates
Hungary	United Kingdom
Iceland	United States
India	Uruguay
Indonesia	Uzbekistan
Iran	Venezuela
Iraq	Vietnam
Ireland	Zambia

Source: GPRD.

Annex II. Event-Study Analysis for Advanced Economies

To complement the event-study analysis in Section VI.B, the same specification for advanced economies was estimated. A joint test of the pre-treatment coefficients (k = −5 to k = −2) yields an F-statistic of 0.36 (p = 0.835), indicating no pre-trend. The average post-treatment effect across k = 0 to k = +5 is −0.86 percentage points (SE = 2.34, p = 0.717), statistically indistinguishable from zero. All individual event-time coefficients are insignificant, with wide confidence intervals spanning zero throughout. These results are consistent with the baseline DiD estimate in Table 4 and confirm that the debt-reducing effect of pension reform is concentrated in non-advanced economies.

Figure A1. Event Study for Advanced Economies



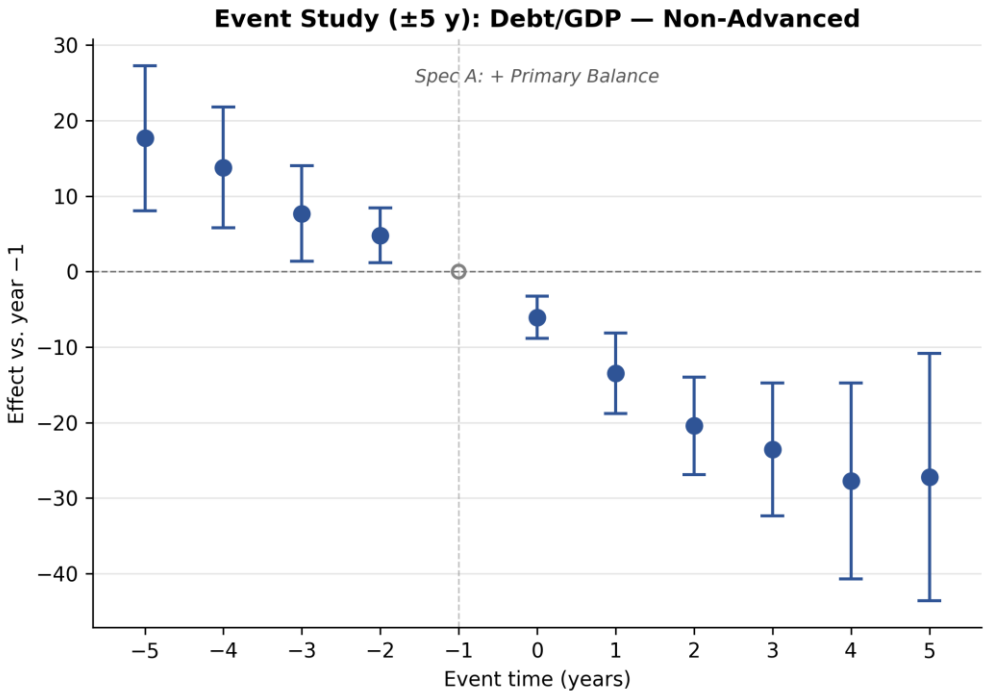
Source: Global Pension Reform Database; and authors' calculations.

Notes: The figure plots event-study coefficients from a TWFE regression of debt-to-GDP on relative-time indicators around the last implemented pension reform for advanced economies. The omitted category is k = −1. Vertical bars represent 95 percent confidence intervals. Standard errors are clustered at the country level. Controls include GDP growth, GDP per capita growth, inflation, the old-age dependency ratio, and a lagged IMF program dummy.

Annex III. Robustness Event Studies

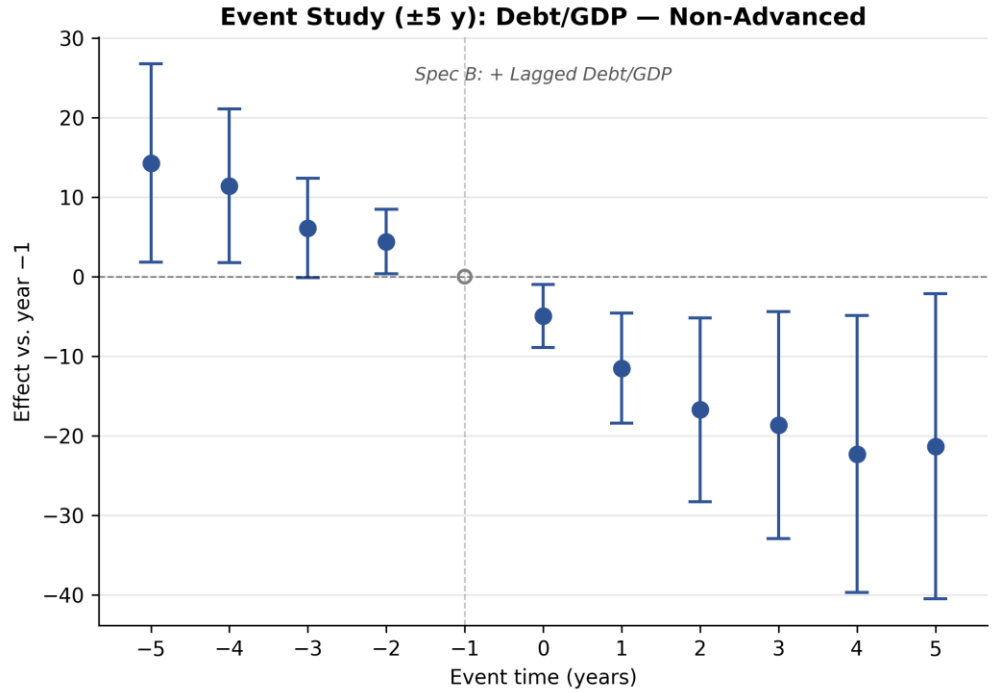
Appendix Figures A2 to A7 present event-study plots for each robustness specification in Table 5. For non-advanced economies (Figure A2 to A4), the dynamic pattern is remarkably stable: all three specifications display a gradual, sustained decline in debt-to-GDP beginning at the year of reform, with no evidence of pre-trending in the lead coefficients. The inclusion of lagged debt-to-GDP in Specifications B and C modestly tightens the confidence intervals without altering the trajectory, suggesting that short-run debt persistence does not confound the estimated reform effect. For advanced economies (Figure A5 to A7), all coefficients remain small and statistically insignificant across specifications. Notably, confidence intervals narrow substantially once lagged debt is included (Specifications B and C), yet the null result persists, reinforcing the conclusion that pension reforms do not generate measurable debt effects in advanced economies.

Figure A2. Event Study: Non-Advanced Economies
Specification A (+ Primary Balance)



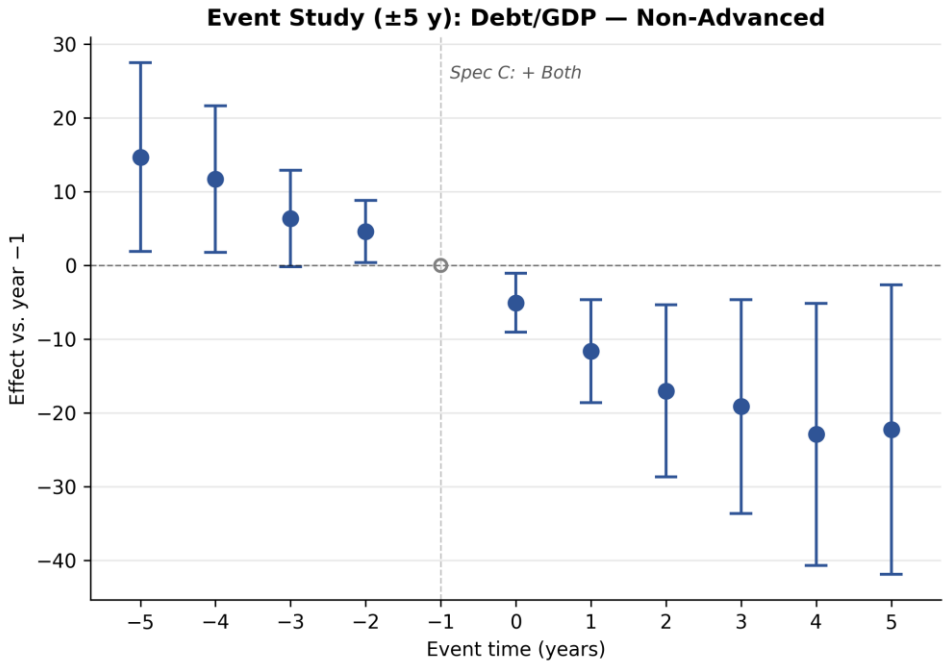
Source: Global Pension Reform Database; and authors' calculations.

Figure A3. Event Study: Non-Advanced Economies
Specification B (+ Lagged Debt/GDP)



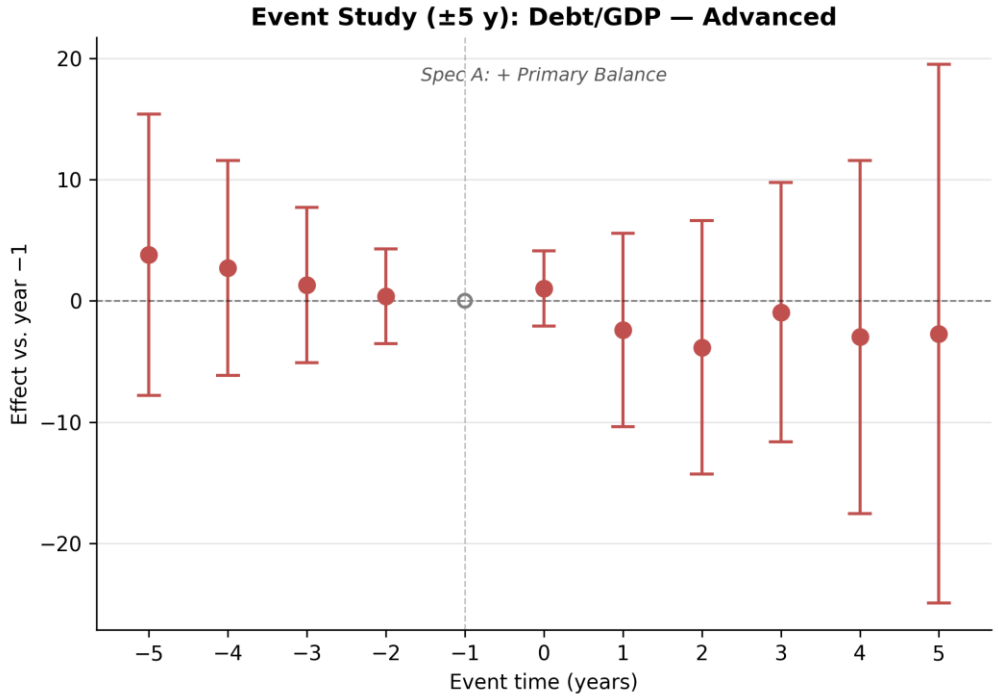
Source: Global Pension Reform Database; and authors' calculations.

Figure A4. Event Study: Non-Advanced Economies
Specification C (+ Primary Balance and Lagged Debt/GDP)



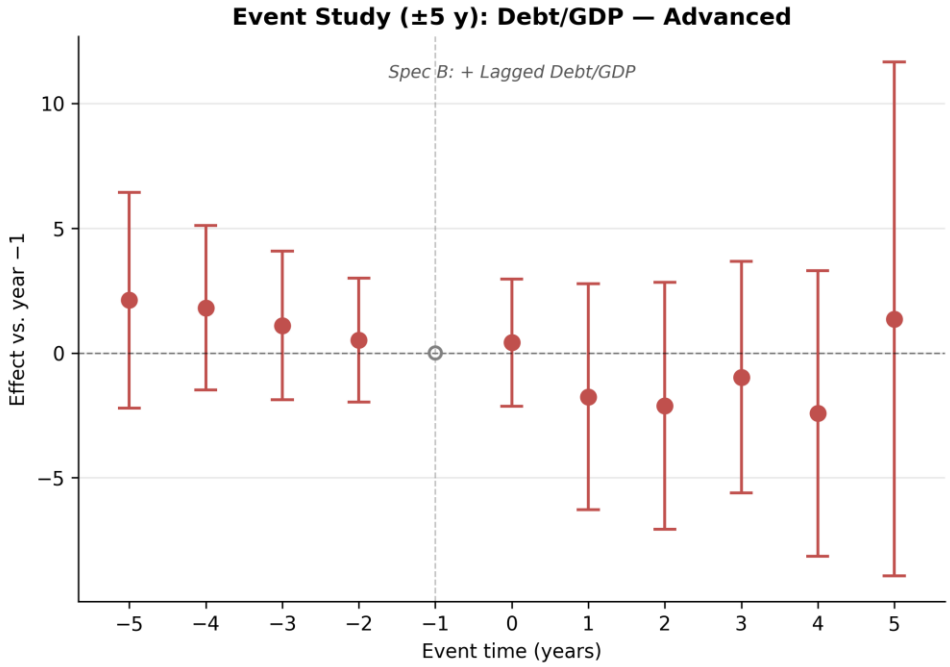
Source: Global Pension Reform Database; and authors' calculations.

Figure A5. Event Study: Advanced Economies
Specification A (+ Primary Balance)



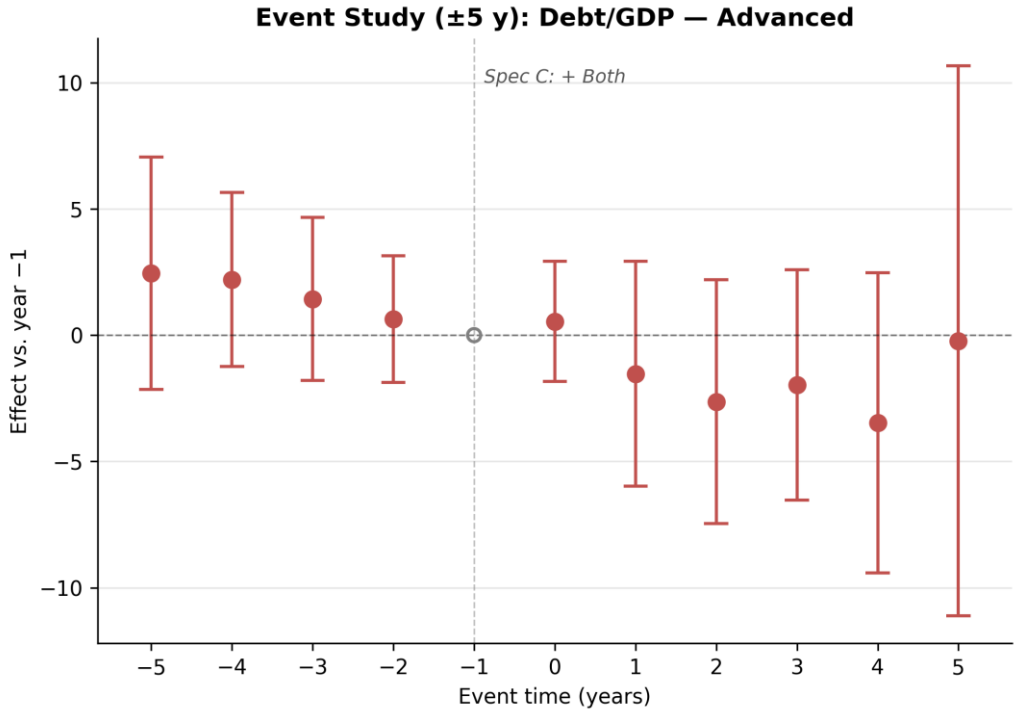
Source: Global Pension Reform Database; and authors' calculations.

Figure A6. Event Study: Advanced Economies
Specification B (+ Lagged Debt/GDP)



Source: Global Pension Reform Database; and authors' calculations.

Figure A7. Event Study: Advanced Economies
Specification C (+ Primary Balance and Lagged Debt/GDP)



Source: Global Pension Reform Database; and authors' calculations.

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